Reviewed Preprint
v1 • December 4, 2025
Not revised

Reviewed Preprint
v2 • April 10, 2026
Revised by authors

Stranded short nascent strand sequencing reveals the topology of DNA replication origins in *Trypanosoma brucei*

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Competing interests: No competing interests declared

Funding: See page 25

Reviewing editor: Yamini Dalal, National Cancer Institute, United States

eLife Assessment

The authors adapt sequencing of nascent DNA (DNA linked to an RNA primer, "SNS-Seq") to map DNA replication origins in Trypanosoma brucei. The main impact of this work is reporting a new set of putative origins, which do not overlap with previously reported origins, but which appear to overlap with previously mapped DNA-RNA hybrid (R-loops). Thus, these **valuable** findings open up new avenues for further investigation into the mechanistic basis for firing of replication forks in this organism. However, the supporting evidence remains **incomplete** and would benefit from orthogonal validation. This work will be of interest to those studying DNA replication and epigenetic regulation of fork origins.

https://doi.org/10.7554/eLife.108143.2.sa3

Abstract

The universal features that define genomic regions acting as replication origins remain unclear. In this study, we mapped a set of origins in *Trypanosoma brucei* using stranded short nascent strand sequencing method. Our results showed that DNA replication predominantly initiates in intergenic regions between poly(dA)- and poly(dT)-enriched sequences. G4 structures were detected in the vicinity of some origins and were embedded in poly(dA)-enriched sequences in a strand-specific manner: G4s on the plus strand were located upstream, while those on the minus strand were located downstream of the centre. The origins' centres were found to be areas of low nucleosome occupancy, surrounded by regions of high nucleosome occupancy. Furthermore, our results demonstrate that 90% of replication origins overlap with a minor proportion of the previously reported RNA: DNA hybrids. These findings shed new light on the sequence and structural features that define the topology of replication origins in *T. brucei*. To further characterize replication dynamics at the single-molecule level, we employed DNA combing analysis.

Introduction

DNA replication is one of the most fundamental processes by which a cell creates an exact copy of its genome prior to division. In eukaryotic cells, DNA replication initiates at multiple genomic sites, which are known as origins. Despite extensive research, a comprehensive universal code that is common for all eukaryotic origins remains elusive (reviewed in [1–3]). The first isolated and characterized eukaryotic origin comes from budding yeast, *Saccharomyces cerevisiae*, where DNA replication starts from the AT-rich autonomously replicating sequence (ARS) with a loosely defined and thymidine-rich ARS consensus sequences (ACS) [4]. The other eukaryotic origins lack a unique consensus sequence. Some studies reported increased AT-content [5–11] or long poly(dA:dT) tracts as replication origins [12,13]. Other reports described the origins as regions of a high GC-content, such as CpG islands [14–17] or regions near G-rich sequences with the potential to form G-quadruplexes (G4s) [18–26]. In addition, several epigenetic factors and chromatin environments were linked to eukaryotic origins, but without a common signature [1,22,27]. However, replication origins exhibit several characteristics that distinguish them from the surrounding chromatin. They appear in nucleosome-depleted regions [28–31], which are flanked on one or both sides by well-positioned nucleosomes [12,29–32]. An active origin is characterized by the generation of two divergent single-stranded DNA molecules, each with a short RNA primer at 5' end, known as short nascent strands (SNS) (reviewed in [33]). Size-selected SNS can be enriched and purified from DNA molecules by treatment with T4 polynucleotide kinase (T4 PNK) and λ-exonuclease, and this approach was developed for mapping of DNA replication initiation sites in yeast [34]. Thereafter, this technique has been successfully applied in different species, using approaches such as SNS-on-chip [15,16] or by high throughput sequencing of SNS (SNS-seq) [18,20,22,25,26,35,36]. A recent development has improved the conventional SNS-seq method by enabling the preservation of SNS molecule directionality following sequencing [17,37].

Trypanosoma brucei is a unicellular parasite of medical and veterinary importance that provokes deadly vector-borne diseases: sleeping sickness in humans, and nagana disease in livestock. This parasite circulates between mammalian and insect hosts, developing various life cycle stages: bloodstream form (BSF) in the mammals and the procyclic form (PCF) in the midgut of the tsetse fly [38]. The order *Trypanosomatida* diverged independently of the *Opisthokonta*, the group comprising animals and fungi [39], and has attracted attention as an original model for biological and comparative studies. These single-celled eukaryotes have some unusual genomic and epigenetic properties. Transcription is polycistronic, meaning that multiple functionally unrelated genes are transcribed into one long precursor mRNA [40]. A better understanding of the fundamental processes in these divergent eukaryotes could provide insight into evolutionary steps and pave the way for innovative medical treatments for sleeping sickness and nagana disease.

The replication of DNA in these parasites has recently become a subject of interest [41–43]. To map and identify active DNA replication origins (hereafter referred to as "origins") in *T. brucei*, we isolated RNA-primed SNS molecules and sequenced them using a stranded protocol that preserves their orientation. This stranded SNS-seq method allowed the mapping of origins between two divergently oriented SNS peaks and the discovery of potential DNA replication start sites. We then conducted an extensive bioinformatics analysis of the mapped origins and compared our findings with previously published scientific datasets. This enabled us to identify several key components of origins and to propose a model of an origin in *T. brucei*.

Results

Origin mapping by stranded SNS-seq

Stranded SNS-seq method was employed to map the origins in two distinct cell types of *T. brucei*, representing two life cycle stages of this parasite: PCF and BSF cells. The experimental workflow for purifying and enriching SNS, together with the corresponding control, is illustrated in Figure 1A. SNS-enriched samples were prepared through a two-stage process. SNS molecules were enriched first, by size-selection (0.5-2.5 kb), and second, by removing broken DNA molecules by λ-exonuclease treatments. SNS-depleted control, prepared in parallel with every SNS-enriched sample, underwent an additional RNase treatment before λ-exonuclease treatment to remove RNA, including RNA primers from the 5' end of the SNS (Methods and Figure 1A). SNS-depleted control contained molecules resistant to RNase and/or λ-exonuclease treatment, representing all molecules that are difficult to digest with lambda exonuclease. This control was utilised as a background control. Then, stranded libraries were prepared for high-throughput sequencing (Figure 1B and Methods). Paired-end sequencing of an SNS-enriched samples and an SNS-depleted controls was performed on three biological replicates of PCF and BSF cells. Read counts after sequencing and each analysis step are shown for each sample in Supplementary Table 1 and Supplementary Figure 1A.

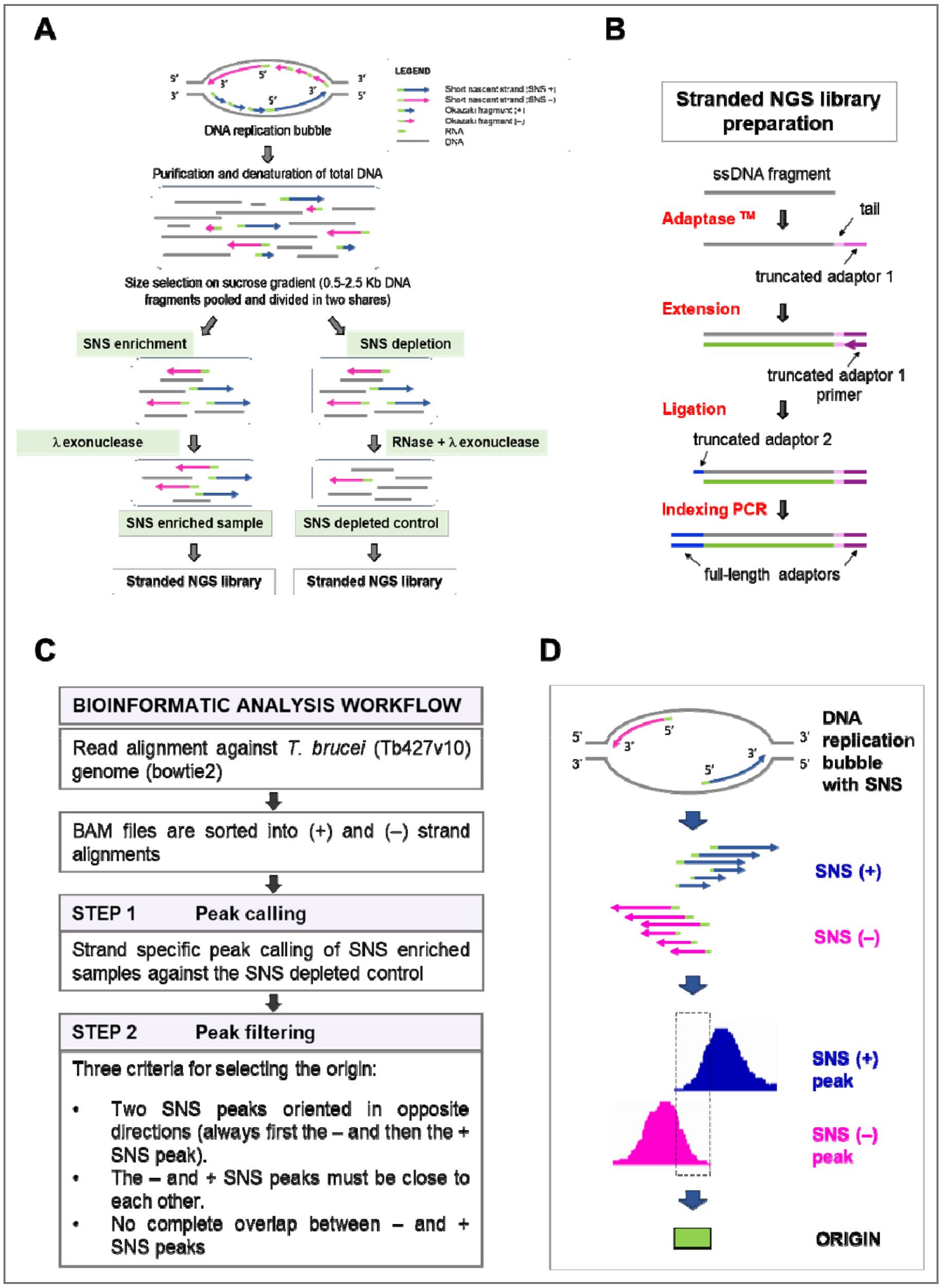

Figure 1. **Experimental protocol and bioinformatics workflow for selecting the origins.**

A) Schematic representation of the experimental workflow for short nascent strand (SNS) purification. A replication bubble with DNA replication intermediates and their orientation is shown at the top. The methodology employed for the generation of the SNS-enriched samples and corresponding SNS-depleted controls is illustrated. B) Workflow of stranded library preparation for Illumina sequencing. Single stranded DNA fragments were tailed and tagged at the 3′ end with truncated adaptor 1. Primer extension was used to generate the complementary strand. The ligation step was used to add the truncated adaptor 2. The indexing PCR step included full-length adaptors. C) Schematic diagram of the bioinformatic workflow applied for stranded SNS-seq data analysis. Two steps of peak selection were performed to map the origins. The first step was peak calling against SNS depleted control. The second step was peak filtering, based on the three indicated criteria. D) The schematic representation shows one origin, generating a replication bubble with two SNS leading strand in divergent orientation (SNS + and SNS –). The plus and minus SNS are presented to illustrate the growth of the leading strand in both directions after the ligation of the upstream Okazaki fragments to the 5′ ends of SNS. The different sizes of the isolated SNS are also presented. The pink peak represents the SNS minus peak synthesized on the plus DNA strand. The blue peak represents the SNS plus peak synthesized on the minus DNA strand. The dashed rectangle, situated between the two inner borders of two divergent SNS peaks, and the green box represent the mapped origin.

The bioinformatics workflow used for origin selection is schematically presented in Figure 1C and Methods. In brief, after the alignment and strand-specific sorting of the reads, two steps of peak selection were performed. The first step was strand-specific peak calling against SNS-depleted control as a background control. The aim of this step was to eliminate false positive peaks resulting from undigested DNA structures, such as G4s and other structures that could withstand λ-exonuclease treatment. The second step, called "peak filtering", identified peaks pairs consistent with the divergent orientation of SNS molecules at active origins (Figure 1C). Therefore, only peak pairs with a minus peak in front of a plus peak, and both in proximity (Methods and Figure 1C and 1D) were retained as positives, while 88-96% of the called peaks were removed in the "peak filtering" step (Supplementary Figure 1B and 1C). Here, we used information about the divergent orientation of SNS, a key feature of active origins, to increase the specificity of origin mapping. Ultimately, the origins were identified as genomic regions situated between the inner borders of filtered peak pairs (Figure 1D).

The stranded SNS-seq data were mapped across the entire genome, including 11 core regions and all non-core regions (subtelomeric and 'unitig')[44]. A representative screenshot of the distribution of plus and minus peaks and the positions of the mapped origins for three replicates of BSF cells is shown in Figure 2A. We found a variable number of origins in the replicates (Supplementary Figure 1D), and the overlap of origins between replicates varied from 8-64%. The distribution of the lengths of the mapped origins was estimated for each replicate (Supplementary Figure 2A), and the average length of origins was found to be approximately 150 bp. After mapping, we merged the origins from the three replicates, identifying a total of 914 origins in BSF cells and 549 in PCF cells. Of the origins in BSF cells, 742 (81.2%) were mapped to the core genomic region and 172 (18.8%) to the non-core regions. In PCF cells, 492 origins (89.6%) were mapped to the core genomic region and 56 (10.4%) to the non-core regions (Figure 2B). Therefore, although these regions present more than half of the genomic sequence (Figure 2B), we mapped fewer origins in non-core regions.

To identify preferred genomic sites of origin activation, we analysed origin distribution around centred intergenic and genic regions. Origins were unevenly distributed, showing higher density in intergenic and lower in genic regions (Figure 2C). This intergenic enrichment was consistent across both cell types and absent in shuffled controls, which consists of randomly selected regions matched for size, number, and chromosomal distribution (Figure 2C). Subsequently, the proportion of origins in four distinct genomic regions was determined, revealing that over 92% of PCF and 85% of BSF origins map within intergenic regions (Figure 2D). This distribution was significantly different from the distribution that would be expected if the origins were distributed evenly across the genome (P-value<0.0001) (Figure 2D).

We then wanted to see if origins were organised as solitary sites or grouped together, so we examined how they clustered within genomic regions of increasing length. Our results showed that *T. brucei* origins were predominantly organised in clusters, with 85.8% of PCF origins and 83.2% of BSF origins organized in clusters of 50 kbp (Supplementary Figure 2B). Empirical cumulative distribution function (ECDF) was calculated for both cell types, and a statistically significant difference was confirmed between the origins and shuffled controls clustering (Figure 2E). The results showed that 50% of the PCF origins were clustered within 11.5 kb regions (shuffled control median was 17 kb), while 50% of the BSF origins were clustered within 6.5 kb regions (shuffled control median was 13 kb) (Figure 2E). This means that origins tend to be activated within initiation zones or clusters rather than as solitary initiation sites in *T. brucei*. This finding is consistent with previous research in other organisms, which has shown that origins are unevenly distributed and tend to form clusters of different lengths[20,22,25]. Next, the intersection of PCF and BSF origins revealed that 238 origins are in common (Supplementary Figure 2C). The number of origins in both cells was compared; for this, the reads were normalised to have an equal number of mapped reads in both cell types (Methods, Read sub-sampling). We showed that BSF cells contained almost twice as many origins as PCF cells (BSF: 844 origins and PCF: 454 origins) (Supplementary Figure 2D).

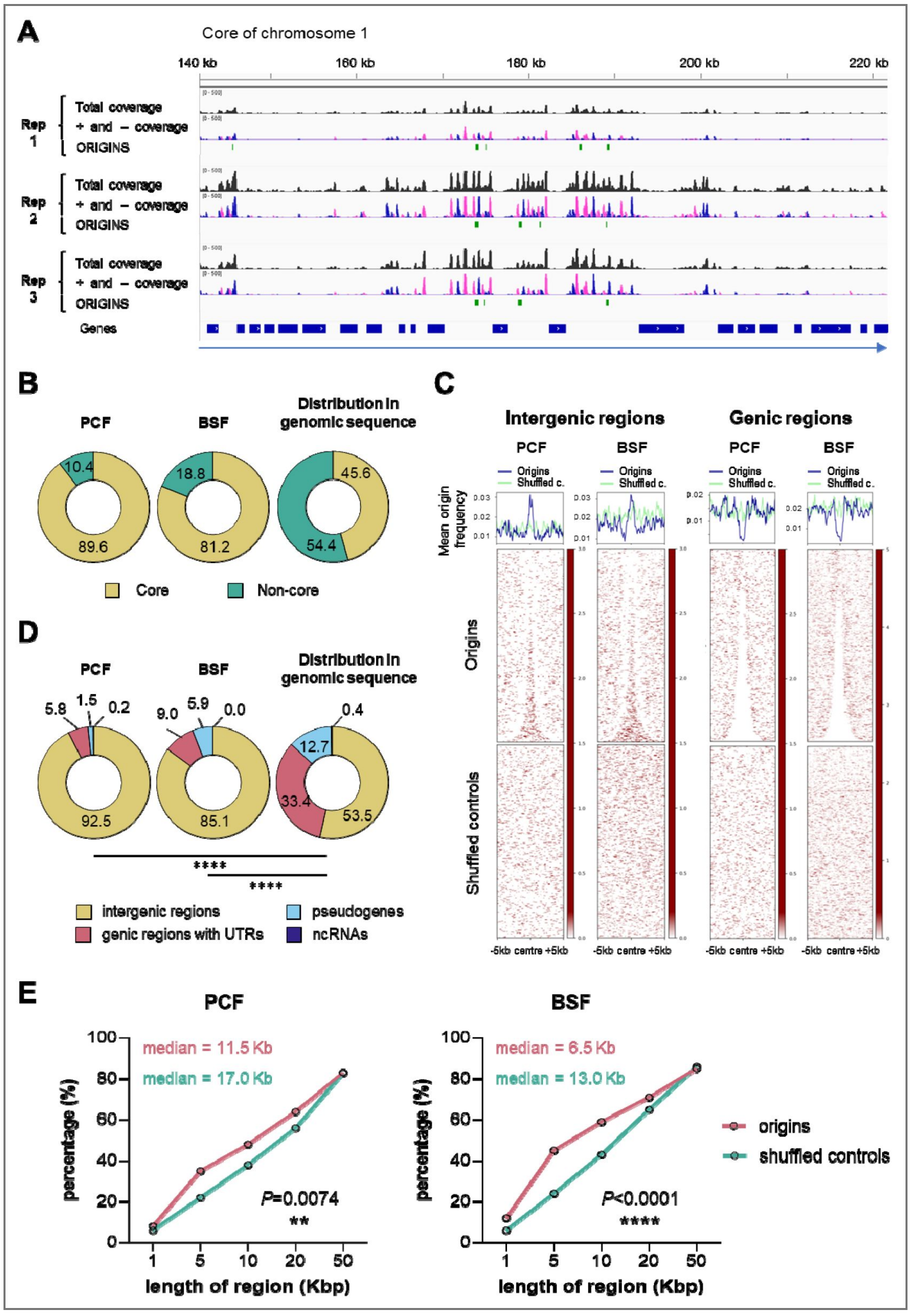

Figure 2. **Genomic distribution of mapped origins.**

A) A representative screenshot from Integrative Genomics Viewer (IGV) presenting the read coverage and the positions of the mapped origins in a ~80 kb long genomic region of the core chromosome 1. The three replicates of BSF cells are presented. The total read coverage is in black, the minus and plus strand read coverage are in pink and blue, respectively. The green bars represent the mapped origins. The blue bars show the position of the genes and the direction of transcription of the polycistronic unit is indicated by the blue arrow below. B) The pie charts show the proportions of mapped origins within the core (yellow) and non-core (green) regions in PCF and BSF cells, as well as the distribution of these two regions within the genomic sequence. C) Heatmaps present the distribution of the mapped origins and shuffled controls within centred genic and intergenic regions (± 5 kb). Shuffled controls present random genomic regions chosen with respect to the size and chromosomal distribution of origins. D) The pie charts illustrate the proportions of the mapped origins of PCF and BSF cells within four indicated genomic regions, as well as the distribution of these four regions within the genomic sequence. The *P*-values were computed using Chi-square tests, which involved a comparison of the absolute numbers of indicated categories for each pair of datasets (**** - *P*<0.0001). D) Empirical cumulative distribution function (ECDF) showing the proportion of pairwise distances between origins (pink) or between shuffled control regions (green). The *P*-values were computed using Chi-square tests.

We then calculated the inter-origin distances (IODs) distribution at the population level for both cell types. The calculated population-based IODs were shorter in BSF (median 15.3 kb) compared to PCF cells (median 21.4 kb) and this difference was statistically significant (Supplementary Figure 2E). Very long IODs were rarely observed, with maximum distances of 354 and 488 kilobases for PCF and BSF origins, respectively (Supplementary Figure 2E).

Insights from DNA molecular combing: comparative analysis of single-molecule DNA replication and cell population origin use

Stranded SNS-seq and DNA molecular combing are two complementary techniques. Stranded SNS-seq provides a genome-wide overview of active origins within a given cell population. However, it should be noted that the usage of these origins may vary between individual cells within the same population (reviewed in [45]). DNA combing provides an overview of DNA replication parameters at the single-molecule level. In this study, we used DNA combing to compare the replication of PCF and BSF cells. Replicating cells were labelled with two modified nucleosides, followed by immuno-detection (Methods). After immuno-detection, the three discrete signals were discerned on combed DNA. The red signal, followed by green, indicated the direction of replication forks, while the blue signal represented the unmodified DNA (Figure 3A). Origins were positioned at the centre of two divergently oriented replication forks. DNA replication parameters, including replication fork speed, IODs, and fork asymmetry, were determined as previously described [41]. Measurements from three biological replicates were pooled and column statistics are shown in Supplementary Table 2. The median replication fork velocity was 1.73 and 2.44 kb/min for PCF and BSF, respectively, while the median IOD was 152 and 213 kb for PCF and BSF, respectively (Supplementary Table 2). The median speed of replication forks was 1.4 times higher, and the median IOD was 1.4 times longer in BSF compared to PCF cells, with these differences between the two cell types being statistically significant (Figure 3B and Figure 3C). Moreover, we compared the frequency of asymmetric replication forks, defined as bidirectional forks that progress from the origin at unequal rates, between the PCF and BSF cell types and found no statistically significant differences (Figure 3D). Given the high sensitivity of DNA combing measurements to the DNA fibre lengths [46], a comparison of the analysed DNA fibre lengths was performed as a control. This analysis showed that there was no statistically significant difference in the length of the analysed DNA molecules between the two cell types (Figure 3E, Supplementary Table 2). This control confirmed that the observed differences in velocity and IODs were not an artefact of the different lengths of the analysed DNA molecules but rather reflected genuine differences between the replication of PCF and BSF cells.

The approximate number of active origins in a single cell can be estimated by dividing the length of genomic sequences (~50 Mb) by the mean or median IODs obtained by DNA combing. It was thus estimated that the range of active origins in a single cell was 279-329 for PCF and 228-235 for BSF cells (Supplementary Table 2). The stranded SNS-seq method revealed 549 origins for PCF and 914 origins for BSF under stringent origin selection, which represent a fraction of all origins. Despite the inability of either method to determine the exact number of origins, the number of origins in a single cell tends to be smaller than in a population of cells. This phenomenon, known as flexible origin usage, is a well-documented feature of eukaryotic origins (reviewed in [45]).

DNA replication starts between poly(dA) and poly(dT) enriched sequences

Having mapped a subset of origins of PCF and BSF cells, our next objective was to identify the common elements that would define the origins of *T. brucei*. To do so, we merged the origins of PCF and BSF cells into a single set, bringing the total number of origins to 1225. All analyses were also performed on the separate PCF and BSF sets of origins to demonstrate the similarity of their structures, and the results are presented in the supplementary figures. As most origins were identified within intergenic regions, another shuffled control was developed to represent random intergenic regions selected in accordance with the number of origins, their size, and chromosomal location (Methods). The mean distance of origins and intergenic shuffled controls from the genes

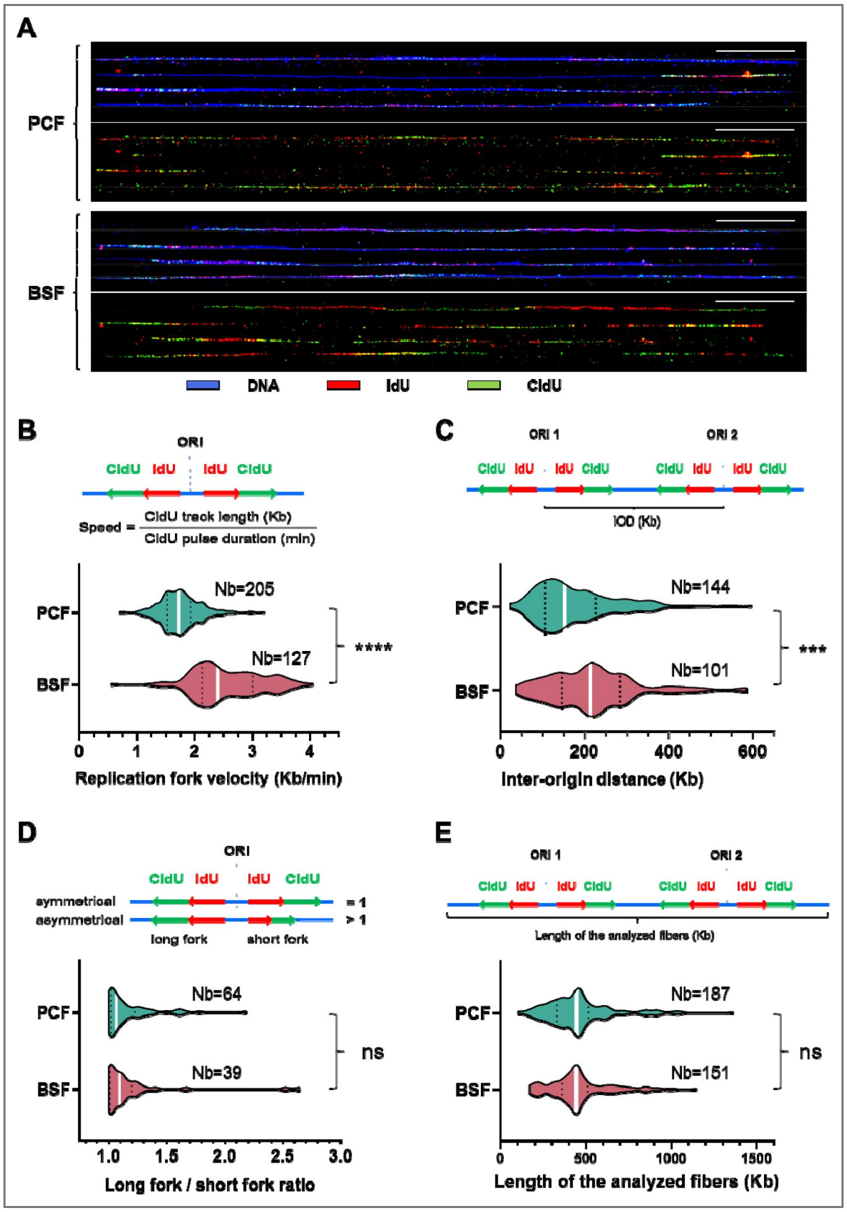

Figure 3. Comparison of DNA replication parameters between PCF and BSF cells by DNA molecular combing.

A) The figure depicts representative DNA molecules after immuno-detection. The immuno-detected DNA is indicated in blue, the initial pulse of nucleosides (IdU) in red, and the subsequent pulse of nucleosides (CldU) in green. The lower panels display only the red and green tracks from the first and second pulse of nucleosides extracted from the corresponding upper panels. 50 kb scale bars are shown as white lines. B) The velocity of the replication forks was calculated by dividing the length of the CldU tracks with the duration of the CldU pulse on intact DNA molecules (schema on upper panel). Violin plots present distribution of the measured replication fork velocities in two cell types. C) The inter-origin distance (IOD) was defined as the length between two adjacent replication initiation sites and can be determined by measuring the centre-to-centre distances between two adjacent progressing forks (schema on upper panel). Violin plots present the distribution of the measured IODs in two cell types. D) The upper panel presents the concept of long fork/short fork ratios. The long fork/short fork ratio corresponds to the ratio of the longer IdU + CldU signal length over the shorter IdU + CldU signal length of bidirectional replication forks. A ratio >1 indicates fork asymmetry, while a ratio =1 indicates fork symmetry [41]. The lower panel presents the distribution of the measured asymmetry of replication forks in two cell types. E) The upper panel presents the concept of the measurements of the analysed DNA lengths. The lower panel presents violin plots with the measured lengths of the analysed DNA in two cell types. White bars on the violin plots indicate median values. Dotted black lines indicate quartiles. Two-tailed Mann-Whitney test was used to compute the corresponding *P*-values (ns - non-significant; * - *P*<0.05; ** - *P*<0.01; *** - *P*<0.001; **** - *P*<0.0001). The number of measurements (Nb) is given for each cell type.

was found to be similar (745.2 bp for origins and 797.6 bp for shuffled controls), thus providing a more accurate control. Our first objective was to examine the nucleotide distribution of centred origins and compared it with the centred shuffled controls. The resulting plots showed that the percentage of adenines (A) or thymines (T) was higher near the centre of the origins (Figure 4A). This percentage was well above 40%, which was higher than the mean A or T content of the intergenic regions (reaching 30%) (Figure 4A), while the overall A or T content of the genomic sequence is approximately 28%. The enrichment near origins showed a remarkable sequence polarity represented by a strong enrichment of A upstream and a strong enrichment of T downstream of the origin centre (Figure 4A, Supplementary Figure 3A). Furthermore, a remarkable polarity in the distribution of guanines (G) and cytosines (C) was also observed. The origins exhibited a slight enrichment of G nucleotides upstream (up to 24%, while the average intergenic G was 20%), coinciding with the A enrichment, and a slight enrichment of the C nucleotides (up to 24%, while the average intergenic C was 20%), coinciding with the T enrichment downstream of the centres (Figure 4A). The nucleotide pattern that was specific for origins was entirely absent in the shuffled controls (Figure 4A, Supplementary Figure 3A). The subsequent step was to search for consensus motives of origins (Methods). However, apart from extended sequences that were enriched with adenines or thymines, no discernible consensus motif was identified (data not shown). The sequences surrounding the origins were also analysed using the WebLogo software [47], and a graphical representation of nucleotide occurrence unambiguously confirmed an unequal nucleotide distribution, with an enrichment of A and G upstream and T and C downstream of the origins (Supplementary Figure 3B). We then looked at the distribution of polynucleotides of different lengths around the origins. Poly(dA)-rich sequences were enriched upstream and poly(dT)-rich sequences downstream of the origins, a pattern absent in shuffled controls (Figure 4B, Supplementary Figure 3C). Interestingly, the upstream region showed an enrichment of up to four consecutive Gs, while the downstream region showed an enrichment of up to four consecutive Cs (Figure 4B). We also checked nucleotide and polynucleotide distributions around PCF and BSF origins separately and found that the distributions were similar in both cell types and equivalent to the merged set of origins (Supplementary Figure 4A and Supplementary Figure 4B). Accordingly, an origin can be defined as a genomic site with a specific uneven distribution of nucleotides, characterised by an overrepresentation of poly(dA)s and poly(dG)s upstream, and poly(dT)s and poly(dC)s downstream of the origin centre.

Subsequently, the proportions of origins that lacked or possessed four As upstream and/or four Ts downstream of the centre were calculated and compared with the intergenic shuffled controls (Figure 4C, left graph). The same calculation was performed for eight As and Ts (Figure 4C, right graph). The findings revealed that 1149 origins exhibited the presence of four As upstream and four Ts downstream of the centre, representing a proportion of ~94% compared to ~78% for the intergenic shuffled controls. Similarly, 450 origins (~37%) possessed eight As upstream and eight Ts downstream of the centre, compared to only 114 (~9%) intergenic shuffled controls with this pattern. Accordingly, the distribution of poly(dA) and poly(dT) observed for origins was found to be significantly different from that observed for the intergenic shuffled controls (Figure 4C).

G4 structures accumulated in a specific pattern in proximity to the origin

G4s, DNA secondary structures with non-canonical Watson-Crick base pairing, have recently emerged as potential regulators of genome replication [48]. Given the previously reported presence of G4 structures in the vicinity of some eukaryotic origins [18,19,21,26], we investigated the correlation between G4s and origins of *T. brucei*. To do this, we used two sets of G4s that had previously been determined by G-quadruplex sequencing (G4-seq) [49]. One set of G4s was obtained under physiological conditions, while the other was obtained in the presence of pyridostatin (PDS), a drug that binds and stabilises G4s [49]. The SNS-seq origin dataset was intersected with stranded G4-seq datasets obtained in both conditions (Methods), resulting in a notable enrichment of G4s in the vicinity of the origins (Figure 5A and Supplementary Figure 5A). A distinctive pattern of G4 distribution was observed, whereby the G4s on the plus strand were found to be enriched

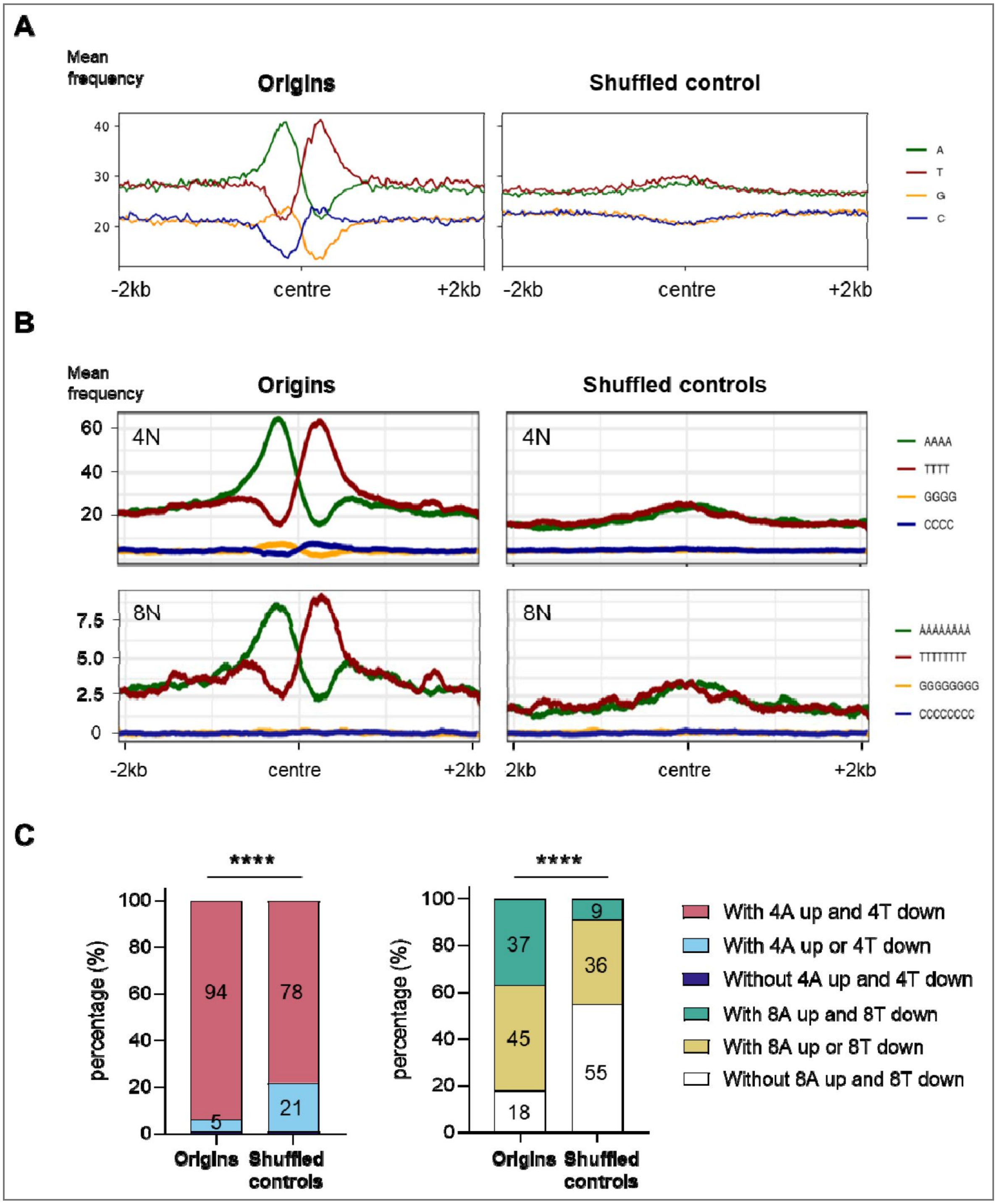

Figure 4. **Spatial organisation of nucleotides and polynucleotides in the vicinity of origins.**

A) The profile plots illustrate the distribution of four nucleotides around centred origins and shuffled controls (± 2 kb). The nucleotide percentage was calculated within 20 bp window. B) The profile plots illustrate the distribution of four and eight polynucleotides around centred origins and shuffled controls (± 2 kb). A smoothing function was employed to calculate the mean frequency of polynucleotides per position within a 100 bp window (50 bp up- and downstream from the position). C) The proportions of origins and shuffled controls without or with four or eight As upstream and/or four or eight Ts downstream of the centre. The ±0.5 kb window from the centre was analysed. The *P*-values were calculated using the Chi-square test, which involved a comparison of the absolute numbers of indicated categories for each pair of datasets (**** - *P*<0.0001).

upstream, while the G4s on the minus strand exhibited an enrichment downstream of the centres of origin (Figure 5A and Supplementary Figure 5A). This pattern of G4 enrichment was absent in the intergenic shuffled controls (Figure 5A and Supplementary Figure 5A). The occurrence of G4s near the origins varied, with a discernible increase of G4s near the centre observed under PDS-stabilised conditions in comparison to physiological conditions (Figure 5A and Supplementary Figure 5A).

Our subsequent objective was to determine the proportion of origins that possessed G4s on one or both sides and those that lacked G4s. To achieve this, the region of ±2 kb around the centres of origins was analysed and compared to shuffled controls. The result showed that the origins displayed significantly different G4 distributions compared to the shuffled controls (P<0.0001) (Figure 5B). Furthermore, the proportions were significantly different in PDS-stabilised compared to physiological conditions (Figure 5B). A set of origins with G4s on both sides constituted a minor population under physiological conditions (9%), whereas this was a major population in PDS-stabilised conditions (39%). The proportion of origins with G4 on one or both sides increased significantly in PDS-stabilised condition (74%), compared to the physiological condition (47%) (Figure 5B). Analyses of the presence of G4s in the region ±0.5 kb around the centres of origins yielded comparable results (Supplementary Figure 5B). Quantification of G4 distribution around the PCF and BSF origins, was similar in the two cell types, and comparable to the merged set of origins (Supplementary Figure 5C).

To gain a deeper insight into the distribution of origins and the G4 structures, an empirical cumulative distribution function (ECDF) was computed. The ECDF demonstrated that the distances between the origins and the closest G4s were smaller than those observed for the shuffled controls (for physiological G4s condition - median of 2010 bp for origins versus median of 4154 bp for intergenic shuffled controls and for stabilised G4s condition – median of 329 bp for origins versus median of 1223 bp for intergenic shuffled controls) (Figure 5C). Half of the origins had a stabilised G4 structure within ±329 bp of the centre, while 80% of the origins had the same structure within ±2575 bp of the centre. Therefore, a strong association was identified between origins and physiological G4 structures (odds ratio = 1.87; P-value = 0), as well as for stabilised G4s structures (odds ratio = 2.13; P-value = 0).

Our results demonstrated that poly(dA) enriched sequences and G4s were enriched upstream of origins on the plus strand, and downstream on the minus strand (Figure 4B and Figure 5A). Therefore, we wanted to determine the spatial relationship between these two elements. To achieve this, the strand separated poly(dA) enriched sequences and the strand separated experimental G4s [49] were intersected at centred origins and shuffled controls (Methods). The results demonstrated that the peaks of the experimental G4s and poly(dA) sequences overlapped near the centres of origins in a strand specific manner that was notably absent in the intergenic shuffled controls (Figure 5D). Our findings revealed a previously unidentified element of origin: the enrichment of poly(dA) and G4 structures exhibited an important association upstream of the centre on the plus strand and downstream of origin on the minus strand (Figure 5D). It was observed that the poly(dA) peaks exhibited summits between ±100 to 300 nt, while the G4 peaks exhibited summits in the range of ±100 to 200 nt from the centres of origins (Figure 5D). Such regions were designated as poly(dA)/G4-enriched elements of origins.

The origins exhibited varying nucleosome occupancy

Previous studies demonstrated the presence of origins in nucleosome-depleted regions [28–31] and their positioning adjacent to well-positioned nucleosomes on one or both sides [12,29–32]. Therefore, our subsequent objective was to determine whether there was a correlation between the origins identified by stranded SNS-seq and the previously published MNase-seq datasets, which described the distribution of nucleosomes in *T. brucei* [50]. After intersecting the origins and nucleosome positioning datasets (Methods), we identified a nucleosome signature specific to origins that was absent in the shuffled control. Origins showed a distinct “M” pattern of nucleosome distribution around the centre, with varying nucleosome occupancy (Figure 6A and Supplementary Figure 6A). The centre of origin displayed a markedly lower nucleosome occupancy (LNO) in comparison

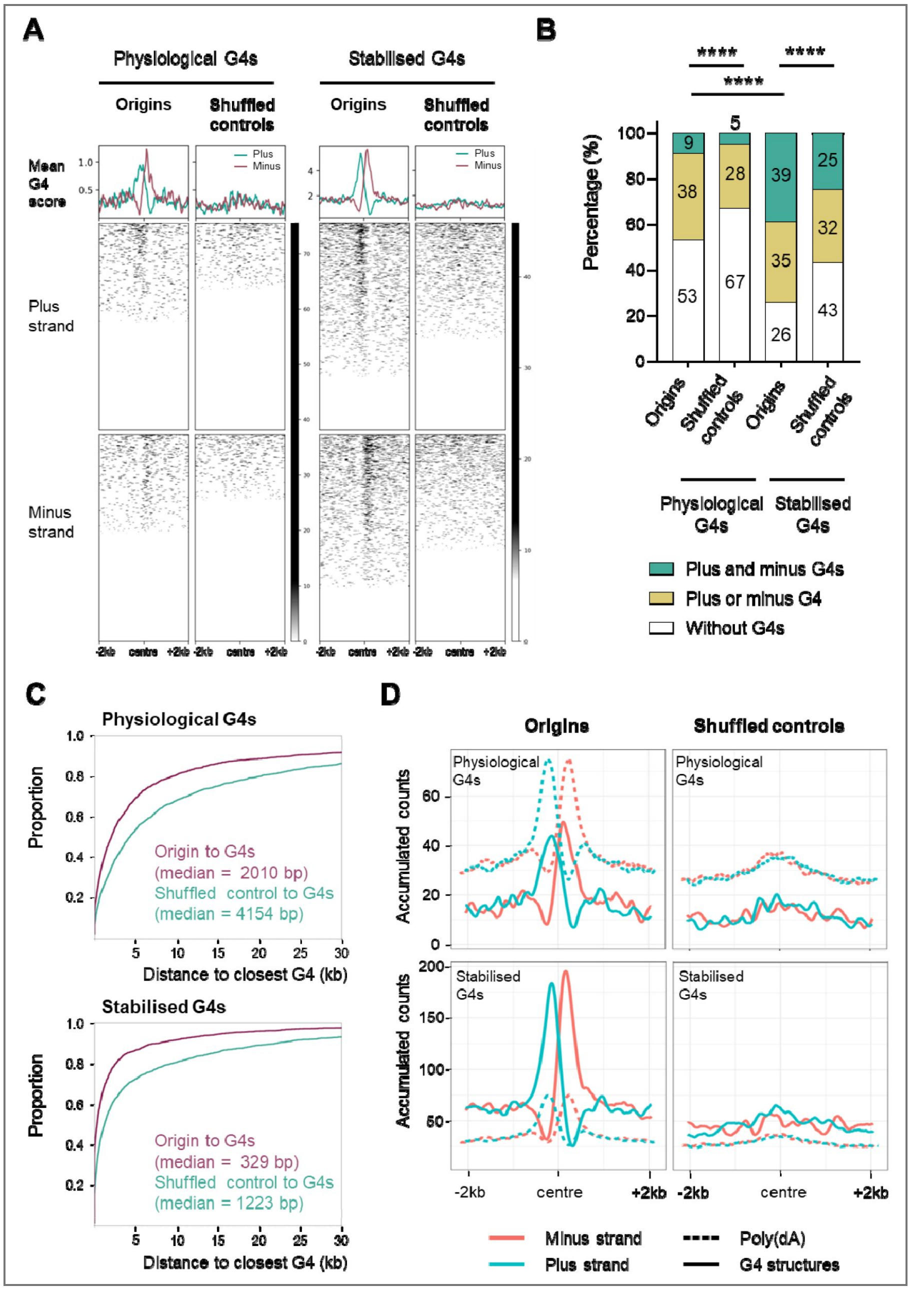

Figure 5. Distribution of G4 structures in the vicinity of origins.

A) The profile plots and heat maps show the distribution of the experimentally obtained G4 structures around centred origins and intergenic shuffled controls (±2 kb) in the Tbb TREU927 reference genome (Methods). The plus strand (light blue) and minus strand (pink) G4s, obtained under physiological conditions and in the PDS drug stabilized condition[49] were overlapped with origins mapped by stranded SNS-seq. Mean G4 score presents average G4 score[49] per 20 bp window. B) The proportions of origins that lack or possess G4 structures on one or both sides of the centre were determined. The ±2 kb window from the centre was analysed for the presence of G4s. Two sets of experimental G4 structures[49] were subjected to analysis in comparison with intergenic shuffled controls. The P-values were calculated using the Chi-square test, which involved a comparison of the absolute numbers of indicated categories for each pair of datasets (**** - P<0.0001). C) Empirical Cumulative Distribution Function (ECDF) of the distances between the origins and the closest physiological and stabilized G4 structures [49]. Median distances are indicated. D) The profile plots illustrate the distribution of the poly(dA) sequence (AAAA) (dashed lines) and the experimental G4s (physiological and PDS drug stabilized)[49] (solid lines) around centred origins and intergenic shuffled controls (±2 kb). The analysis was conducted in the Tbb TREU927 reference genome (Methods). The plus strand is represented in blue and the minus strand in pink. A smoothing function was employed to calculate the accumulated counts of the AAAA and G4s per position within a 100 bp window (50 bp up- and downstream from the position).

to the adjacent regions. Additionally, the centre of origin was situated between two regions exhibiting a higher nucleosome occupancy (HNO) in comparison to the shuffled controls (Figure 6A and Supplementary Figure 6A).

We then quantified the number of origins without or with HNO regions on one or both sides of the centre, and the presence or absence of LNO in the centre and compared them to the intergenic shuffled controls (Methods). We performed this quantification for the merged set of origins (Figure 6B), but also for the mapped origins in PCF and BSF cells against four replicates of the PCF and four replicates of the BSF MNase-seq datasets [50] (Supplementary Figure 6B). Six categories of different combinations of HNO and LNO were analysed, and quantification showed that there were statistically significant differences between origins and corresponding shuffled controls when PCF and BSF origins were merged (Figure 6B) and when PCF and BSF origins were analysed separately (Supplementary Figure 6B). We then analysed each of the six categories individually to determine which had changed significantly between the origins and the shuffled controls. It was found that three categories were significantly different between origins and shuffled controls, while three categories were not (Figure 6C). The category two HNO flanking LNO was significantly enriched (twice more), whereas the category no HNO with LNO and the category no HNO no LNO were significantly depleted in the origins compared to the shuffled controls (Figure 6C). Therefore, the active origins had significantly more LNO flanked by two HNO, at the expense of the no HNO with LNO and the no HNO no LNO categories.

Further investigation was undertaken to determine the spatial relationship between poly(dA)/G4 enriched sequences and nucleosome distribution. To achieve this, strand separated poly(dA) enriched sequences and predicted G4s were overlapped with the MNase-seq data [50] (Methods). The plot obtained after the intersections revealed that the LNO region was positioned between the plus and minus poly(dA)/G4 summits (Figure 6D). Additionally, the HNO peaks were found to exhibit overlap with the poly(dA) and G4 peaks and to surround the origins (Figure 6D). Furthermore, these intersections permitted the estimation of the averaged distance of the above-mentioned elements from the centre of origins. The LNO region extended to approximately ±100 to 150 nt from the centre, thereby defining an LNO area of approximately 200 to 300 bp in the centre of origin. The two HNO regions were located between 150 and 500 nt upstream and downstream from the centre of origin (Figure 6D).

The origins colocalized with a subset of DRIP-seq enrichment sites

DRIP-seq is a widely used technique for genome-wide mapping of RNA:DNA hybrids. This method is based on the immuno-precipitation of fragmented genomic DNA by the S9.6 monoclonal antibody, which specifically recognises RNA:DNA hybrids [52,53]. A recent study employed DRIP-seq in *T. brucei* BSF cells, revealing enrichment within intergenic regions, between coding sequences of polycistronic transcription units, and coinciding with sites of polyadenylation and nucleosome depletion [54]. Considering the established role of short RNA primers in initiating DNA synthesis at the leading and lagging strands, an examination of the colocalization of the origins and previously published DRIP-seq data [54] was conducted (Methods). A significant accumulation of RNA:DNA hybrids was identified around the origins, with two distinct summits located close to the origin centre (Figure 7A). An enrichment was also evident in the vicinity of the centre of the intergenic shuffled controls (Figure 7A), which is consistent with previous reports indicating a high prevalence of RNA:DNA hybrids (R-loops) in intergenic regions [54]. However, this peak was observed to be lower in comparison to the peak observed near centres of origins. Furthermore, the two upper summits were absent in the shuffled controls (Figure 7A). Next, we quantified, the number of origins and shuffled controls that overlapped with R-loops [54]. The results demonstrated that 90% of the origins and 68% of the shuffled controls exhibited overlap with the DRIP-seq enrichment (Figure 7B); a statistically significant correlation was observed (odds ratio = 4.17; *P*-value 2.87E^{-142}), suggesting a link between RNA:DNA hybrid formation and origin activity. As DRIP-seq was only performed on BSF cells [54], we intersected this set of data with PCF and BSF origins separately and found that the distribution of RNA:DNA hybrids around BSF origins was equivalent to the merged set of origins (Supplementary Figure 7A). The distribution of RNA:DNA

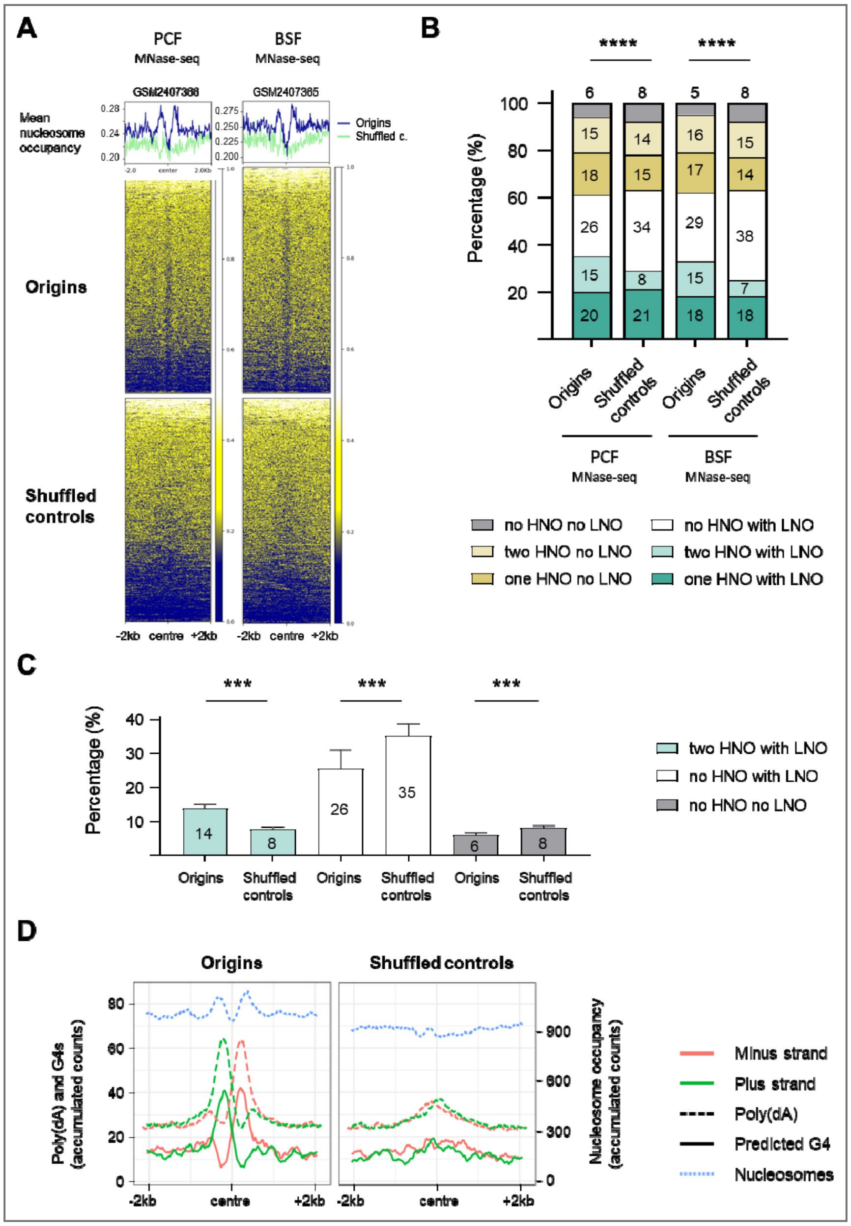

Figure 6. The distribution of nucleosomes around the origins in T. brucei.

A) The profile plot and heatmaps show the distribution of nucleosomes (MNase-seq data) detected in PCF and BSF cells[50] around the centred origins (blue line) and the intergenic shuffled controls (green line) (± 2 kb) in Tb Lister 427 reference genome. One replicate of MNase-seq data[50] for PCF (GSM2407366) and one replicate for BSF cells (GSM2407365) is shown here, the other replicates are presented in Supplementary Figure 6A. Mean nucleosome occupancy presents average nucleosome score (dyad value)[50] per 20 bp window. B) The percentage of origins and intergenic shuffled controls with or without high nucleosome occupancy (HNO) and low nucleosome occupancy (LNO) regions in the indicated combinations. Quantification was performed for the same replicates as indicated in the Figure 6A. The *P*-values were calculated using the Chi-square test, which involved a comparison of the absolute numbers of indicated categories for each pair of datasets (**** - *P*<0.0001). C) Percentage of the three specified categories of HNO and LNO combinations that were significantly different between origins and shuffled controls. The remaining three categories were not statistically significant. The numbers indicate mean percentage values. The bars indicate standard deviations. Mann Whitney two-tailed test was performed to calculate *P*-values (*** - *P*<0.001). D) The profile plots show the distribution of the poly(dA) sequence (AAAA) (dashed line), the predicted G4 structures (solid line) and the nucleosome distribution of the replicate GSM2407365 from BSF cells[50] (blue dotted line) around centred origins and intergenic shuffled controls (± 2 kb). The G4s were predicted by the G4Hunter application in the Tb Lister 427 reference genome. The plus strand is represented in green, and the minus strand in pink. The values on the plots present the accumulated counts of the AAAA, G4s and nucleosome occupancy scores per position within a 100 bp window (50 bp up- and downstream from the position).

hybrids was also enriched around the centres of PCF origins with one of the upper summits less pronounced (Supplementary Figure 7A). Quantification of SNS-seq and DRIP-seq overlap was performed and it was shown that 86% of PCF origins and 93% of BSF origins overlapped with R-loops (Supplementary Figure 7B). It is worth noting that these intersections also revealed that only a small proportion of the previously identified RNA:DNA hybrids (R-loops) [54] overlapped with the merged origins (1.7%), PCF origins (0.7%), or BSF origins (1.3%).

Subsequently, the predicted G4 structures and poly(dA) enriched sequences from the plus and minus strands were superimposed with the DRIP-seq dataset [54] at the centres of origins and shuffled controls (Figure 7C). This intersection allowed us to estimate the relative positions of these elements and their averaged distance from the centre of origins. The two distinct summits of R-loops were observed to be situated, on average, between the centre and 200 nt upstream and downstream from the centre of origins (Figure 7C).

Comparison of the origins identified using stranded SNS-seq with previously published MFA-seq and TbORC1/CDC6 binding sites datasets

We compared the replication origins identified by stranded SNS-seq with the data obtained by Marker frequency analysis coupled with deep sequencing (MFA-seq) [42,43]. MFA-seq compares read coverage between replicating and non-replicating cells and identifies regions with increased DNA coverage in S phase, representing replicated regions. In *T. brucei*, MFA-seq identified broad regions of a few hundred kilobases [42,43], whereas the origins detected by stranded SNS-seq are about 150 bp long. We used six MFA-seq datasets [42,43] (Methods) and we presented core chromosomal overview of the MFA-seq regions [42,43], TbORC1/CDC6 binding sites [42] and the stranded SNS-seq origins on the 11 core chromosomes of *T. brucei* (Supplementary Figure 8A). We found that 28-42% of the origins identified by stranded SNS-seq overlapped with early and 43-55% overlapped with late S-phase MFA-seq replicated regions (Supplementary Figure 8B). We also reanalysed the overlap between TbORC1/CDC6 binding sites [42] and MFA-seq regions [42,43]. Our findings indicated that this overlap is in a similar range as the overlap between SNS-seq origins and MFA-seq replicated regions (Supplementary Figure 8B).

Several proteins have been identified as potential subunits of the origin recognition complex (ORC) in *T. brucei*. One of these is TbORC1/CDC6, which complemented the yeast protein Cdc6, but not Orc1, in a phenotypic complementation assay [55]. We intersected the origins detected by stranded SNS-seq with the previously detected binding sites of the TbORC1/CDC6-12Myc tagged protein [42] (Methods). For this intersection, we used the data from TriTrypDB that contains 6482 ORC1/CDC6 binding sites, we filtered this dataset at score 22 to obtain 990 binding sites which is close to the 953 TbORC1/CDC6 binding sites described previously [42] (Methods). The intersection revealed that TbORC1/CDC6 levels are lowest at the centre of origins (Supplementary Figure 8C). We observed that 12.2% of SNS-seq origins overlap with 24.2% of TbORC1/CDC6 binding sites within a ±2 kb window, 17% of SNS-seq origins overlap with 24.2% of ORC1/CDC6 binding sites within a ±3 kb window and 26.9% of SNS-seq origins overlap with 53.4% of ORC1/CDC6 binding sites in ±5 kb window. Therefore, we identified a depletion of TbORC1/CDC6 binding sites at the centre of the origins, accompanied by a slight accumulation of ORC1/CDC6 binding sites 5 kb upstream and downstream of the origins (Supplementary Figure 8C).

Spatial coordination between the activity of the origins and transcription

DNA replication in trypanosomatids occurs in a particularly challenging environment, as most of their genomes are constitutively transcribed. Arrays of genes oriented in the same direction are transcribed by RNA polymerase II into polycistronic transcription units (PTUs), which can contain hundreds of genes (reviewed in [40]). The primary transcript is processed into individual mRNAs by trans-splicing of a capped 'spliced leader' at the splice acceptor site (SAS) and by polyadenylation at the polyadenylation sites (PAS) (reviewed in [40]). To gain insight into the relationship between

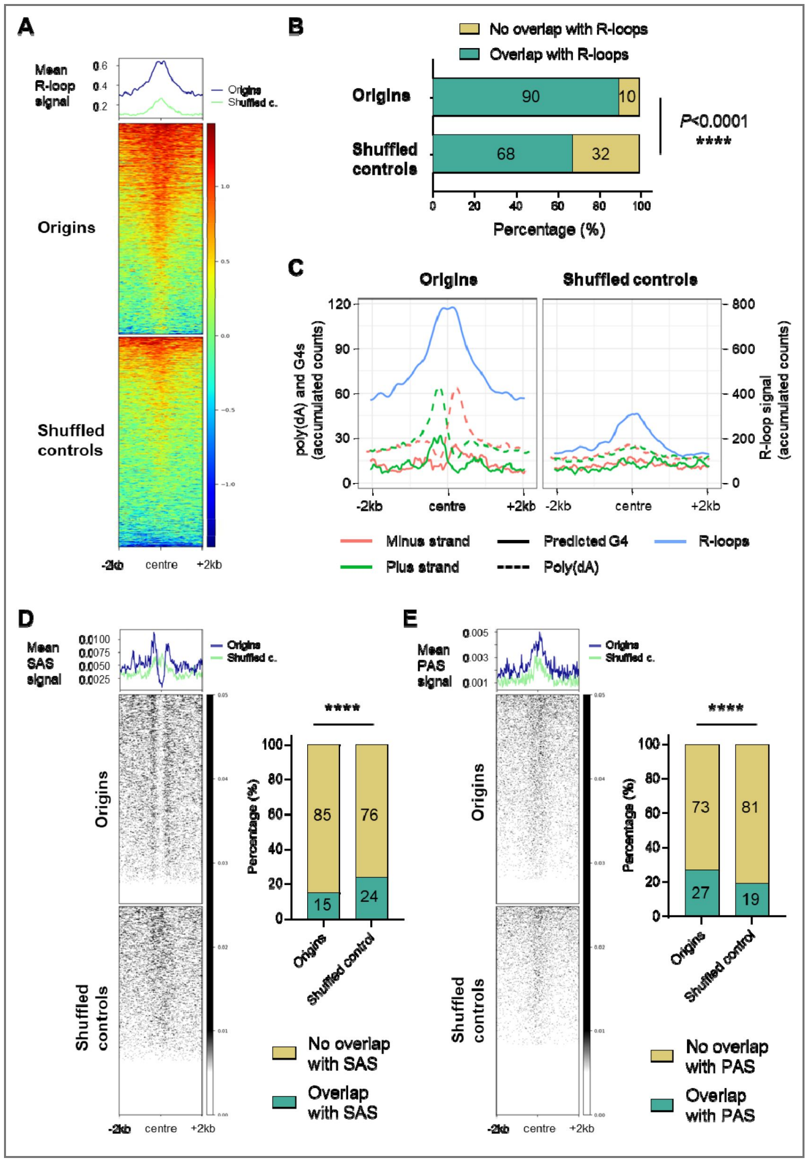

Figure 7. The distribution of DRIP-seq enrichment, splice acceptor sites (SAS) and polyadenylation sites (PAS) around the origins.

A) The profile plots and heatmaps show the distribution of R-loops[54] around the centred origins (blue line) and the intergenic shuffled controls (green line) (± 2 kb). Mean DRIP-seq signal presents average DRIP-seq signal[54] per 20 bp window. B) The proportion of origins and shuffled controls that overlap with R-loops (intersection without window). The *P*-values were calculated using Fisher's exact (two-sided) test, which involved a comparison of the numbers of indicated categories for each pair of datasets (**** - P<0.0001). C) The profile plots illustrate the distribution of the stranded poly(dA) sequence (AAAA) (dashed line), predicted G4s (solid line) and R-loops[54] (solid blue line) around centred origins and shuffled controls (± 2 kb). The plus strand is represented in green, and the minus strand in pink. The 13,409 G4s were predicted by the G4Hunter application[51] in the Tb Lister 427-2018 reference genome. The values on the plots present the accumulated counts of the AAAA, G4s and R-loop signals per position within a 100 bp window (50 bp up- and downstream from the position). D) Left panel: the profile plots and heatmaps show the distribution of transcription splice acceptor sites (SAS) around the centred origins (blue line) and the intergenic shuffled controls (green line) (± 2 kb). Intersection was performed in the Tbb TREU927 reference genome for the 11 megabase chromosomes. Right panel: the proportion of origins and shuffled controls that overlap with SAS (intersection without window). The *P*-value was calculated using Fisher's exact (two-sided) test, which involved a comparison of the numbers of indicated categories for each pair of datasets (**** - P<0.0001). E) Left panel: the profile plots and heatmaps show the distribution of transcription polyadenylation sites (PAS) around the centred origins (blue line) and the intergenic shuffled controls (green line) (± 2 kb). Intersection was performed as in Figure 7D. Right panel: the proportion of origins and shuffled controls that overlap with PAS. The *P*-values were calculated using Fisher's exact (two-sided) test, which involved a comparison of the numbers of indicated categories for each pair of datasets (**** - P<0.0001).

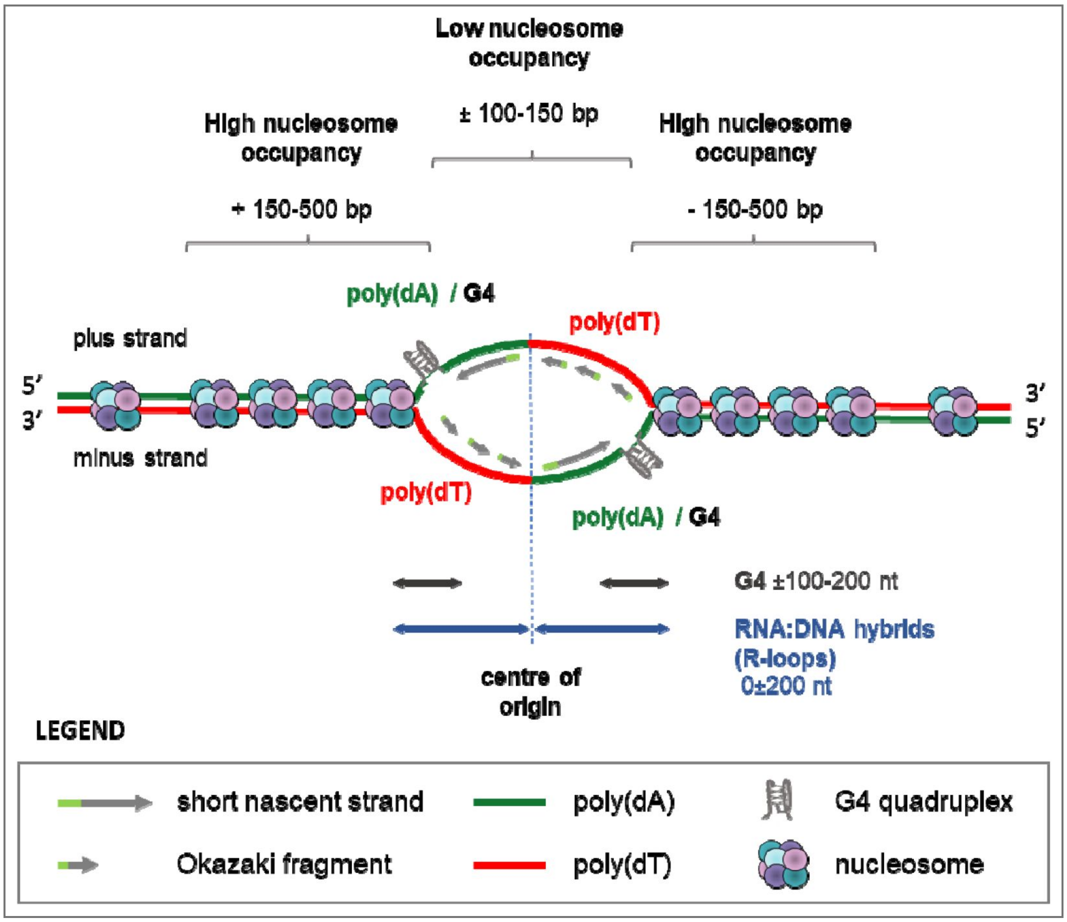

Figure 8. **A proposed model of origin with the position of different genetic elements and nucleosome occupancy.**

Poly(dA) enriched sequences, interspersed with G4 (poly(dA)/G4) are enriched upstream of the origin centres on plus strand and downstream of origin centres on minus strand. Poly(dT) enriched sequences are enriched downstream of origin centres on plus strand and upstream on minus strand. The centre of the origin is a low nucleosome occupancy (LNO) region, flanked by high nucleosome occupancy (HNO) regions. The double arrow lines indicate the position of the summits of the peaks of different origin elements. The presented positions were calculated from the averaged distances. It should be noted that not all origins have the same spacing. The G4 structures, LNO and HNO regions were identified at a limited number of origins; however, they are illustrated on this model to demonstrate the potential of origins to form these structures.

origins and SAS and PAS, we intersected our dataset with the unified SAS and PAS datasets for the core chromosomes (Methods). The intersection of origins with SAS showed that the centre of origins was depleted of SAS, while some origins had SAS flanking the centre and the SAS signal was enriched ±200-500 nt from the centre of origins (Figure 7D, left panel). Only 15% of the origins overlapped with SAS compared to 24% of the shuffled controls, and this difference was statistically significant (Figure 7D, right panel). The intersection of the origins with PAS revealed an accumulation of PAS near the origin centres. A similar, though less pronounced, accumulation was also observed in the shuffled controls (Figure 7E, left panel). Quantification showed that 27% of origins overlapped with PAS compared to 19% of shuffled controls and this difference was statistically significant (Figure 7E, right panel).

PTUs are separated by strand-switching regions (SSRs) and head-to-tail (HT) regions. SSRs are classified as either divergent (dSSRs) or convergent (cSSRs). dSSRs and cSSRs serve as transcription start (TSS) and termination sites (TTS), respectively (reviewed in [40]). Our results showed that the origins are predominantly located in the intergenic regions (Figure 2C and 2D). The metaplots revealed that the TSS, TTS and HT regions [56] were not enriched with origins identified by stranded SNS-seq (Supplementary Figure 8D). Only 3.7% of origins are located within 30% of TSS, TTS and HT regions. Therefore, most origins are localized in actively transcribed regions, which could lead to collisions between DNA replication and the transcription machinery. This spatial coincidence implies that transcription and replication must occur in a highly ordered and cooperative manner in *T. brucei*.

Discussion

We used stranded SNS-seq to identify a set of active origins in the unicellular eukaryote, *T. brucei*. Our methodology is comparable to previously published origin-derived single-stranded DNA sequencing (Ori-SSDS) [17] but includes few differences. The stranded library preparation process was different in these two approaches, and the SNS-depleted control was included in the origin mapping process as a background control. The key benefit of stranded over conventional (non-stranded) SNS-seq is the ability to identify origins with greater specificity by eliminating peaks that do not align with the proper orientation of SNS of an active origin. In addition, this method maps the origins between two divergent SNS peaks, allowing the determination of the centre of origins or replication initiation sites, which was essential for the discovery of the positions and orientations of the elements involved in the topology of origins. Origins mapped between two divergent SNS peaks showed the origin-specific nucleotide distribution around the centre (Figure 4A, Supplementary Figure 9A). On the other hand, if we apply conventional SNS-seq origin mapping to our data and map origins in the middle of peaks without knowing their orientation, the origin-specific nucleotide distribution is lost (Supplementary Figure 9B). Therefore, the primary benefits of our approach include the ability to centre origins between two divergent SNS peaks and to eliminate false positive peaks.

Using this stringent origin selection procedure, we have selected and analysed a subset of all origins. Some weak or inefficient origins may be missed due to low activity, resulting in low read coverage. It is therefore very likely that the number of origins of this parasite is higher than the number reported here. We also examined the origins separately in the two life stages demonstrating that the origin structures in the PCF and BSF cells were equivalent and justifying the merging of these two sets of origins into one. Under this stringent origin mapping condition, we found that 238 origins are in common (Supplementary Figure 2C). However, we decided not to compare the PCF and BSF sets of origins in more detail because comparing non-exhaustively mapped origins in these two life cycle stages could be misleading rather than indicative of true differences between the two cell types.

Our analysis revealed no consensus sequence associated with the origins, which is consistent with the findings of most other studies of eukaryotic origins, including trypanosomatids [22,27,42,43]. Nevertheless, the genomic sites that function as origins are not random sequences. Thanks to the stranded SNS-seq approach, we were able to determine the positions and strand-specificity of some genetic elements and to present a comprehensive model of an origin (Figure 8). The results of our study showed that DNA replication starts at genomic sites where adenine and thymine are overrepresented and arranged in a specific pattern. On the plus strand, poly(dA) enriched sequences were situated upstream, while poly(dT) enriched sequences were located downstream of the origin's centre (Figure 4B and Figure 8). The G4s near the centre of origins were embedded in poly(dA) rich sequences and the summits of G4 peaks covered 100-200 nt up and downstream from the centres (Figure 5D and Figure 8). Our model shows that SNS molecules are synthesised using poly(dA)/G4-rich complementary strands as templates (Figure 8). In our study the enriched molecules detected in divergent peaks do not contain G4s. Rather, they are complementary to the strands enriched with G4s. Therefore, the observed enrichment of G4s around origins cannot be an artefact of the enzymatic treatments, as previously suggested [35]. The centres of origin were identified as LNO regions, which had an average length of 200–300 nt and were flanked by HNO regions located from ±150 to ±500 bp from the centre (Figure 6D, Figure 8). RNA:DNA hybrid enrichment revealed two distinct summits, one upstream and one downstream of the centre, with a range spanning ±200 nucleotides (Figure 7A, 7C and Figure 8).

Our analysis indicated that G-rich sequences near origins tend to form G4 structures (Figure 5A-C, Supplementary Figure 5B-C). The detection of more G4s under PDS-stabilized conditions compared to physiological ones suggests that there is an important potential for G4s formation near origins. G4 formation seems to be very transient and therefore not easy to detect unless stabilized by drug. It seems that G4 structures near origins undergo rapid changes, either forming or dissolving in response to the replication dynamics. We therefore propose that the formation of G4s plays a regulatory role in the activation of replication origins. Some of the recent publications suggested that G4s may serve as replication fork barriers and play a role in transient replication fork stalling [21,57]. The enrichment of G4 structures on strands acting as templates for leading strand synthesis implies that the formation of G4 structures could potentially pose or arrest replication fork progression.

Our analysis disclosed for the first time a strong link between RNA:DNA hybrid formation and DNA replication initiation. Given that RNA-primed nascent strands naturally form RNA:DNA hybrids during replication initiation, the enrichment of RNA:DNA hybrids near origins is both expected and biologically relevant. We found that a small proportion of previously reported RNA:DNA hybrids (1.7%) overlap with SNS-seq origins and we suggest that these RNA:DNA hybrids represent the SNS-derived RNA:DNA hybrids et the DNA replication initiation sites. The S9.6 antibody used for immunoprecipitation in DRIP-seq experiments exhibits a high affinity for RNA:DNA hybrids longer than 6 bp [58]. The RNA:DNA hybrids are formed at the replication initiation sites by RNA priming of the SNS and Okazaki fragments and S9.6 antibody can immunoprecipitate these RNA:DNA hybrids. However, Okazaki fragments are not exclusive markers of origins as they form RNA:DNA hybrids throughout the genome, whereas RNA:DNA hybrids formed by RNA-primed SNS are specific markers of origins. Therefore, we suggest that the summits of RNA:DNA hybrids near the origins reflect the enrichment of RNA-primed SNS molecules. We also consider that the overlap between the two distinct techniques, DRIP-seq and stranded SNS-seq, provides compelling validation of our origin mapping approach.

The elements presented in our model were not found in all origins, as revealed by the quantification performed in this study. This may be caused by the fact that the datasets used for the intersections were obtained from four different strains, rather than from just one. Different strains are likely to activate different sets of origins, which could introduce bias during the intersection process. Another possible source of bias may be the varying quality of the genome assemblies, since this study used three different reference genomic sequences (Methods). This is clearly demonstrated by the detection of different numbers of origins in different reference genomic sequences using the same stranded SNS-seq data (Methods). An alternative hypothesis to explain the absence of certain structures in some origins is that they form transiently, only during origin activation. Therefore, it can be postulated that the presence of most of the elements, in addition to the chromatin state and nuclear organisation of origins, defines the moment of origin activation.

Different origin mapping methods often demonstrate limited overlap due to biases related to the chosen methodology [59,60]. SNS-seq and MFA-seq are two complementary techniques that provide information on different aspects of DNA replication. SNS-seq is a widely used method of high-resolution origin mapping, whereas MFA-seq identifies replicated genomic regions in synchronized cells by comparing the number of DNA copies in the S phase and other phases of the cell cycle [60,61]. The broad replicated regions (0.1–0.5 Mbp) identified by MFA-seq in *T. brucei* [42] are likely to contain multiple origins, rather than just one. In this regard, we disagree with the interpretation of McCulloch's group, who postulated that there is a single origin per broad peak [42]. Our analysis shows that up to 28-55% of the origins detected by stranded SNS-seq locate within MFA-seq regions, providing complementary information on the position of origins within broad replicated regions. It should also be noted that the experimental context was different: MFA-seq relies on cell sorting, while stranded SNS-seq is performed on asynchronous cell populations, enabling the identification of origins active throughout S phase. A similar result was found when comparing conventional SNS-seq and MFA-seq in another trypanosomatid, *Leishmania major* [22].

We found a minority of TbORC1/CDC6 binding sites near origins detected by SNS-seq and this was comparable with the previously reported ~4.4% overlap of this protein with MFA-seq data (reviewed in [62]). Comparable results were also obtained in *T. cruzi* [27]. DNAscent nanopore sequencing revealed 4,287 origins in *T. cruzi*. Of these, 3,782 (88.2%) did not match the 851 TcORC1/CDC6-Ty1 binding sites within a ±3 kb window [27]. Therefore, only 11.7% of origins of *T. cruzi* overlapped with TcORC1/CDC6-Ty1 binding sites within a ±3 kb window, a proportion similar to what we observed in *T. brucei* (6.2%). The origins detected by conventional SNS-seq in *Plasmodium falciparum* revealed 63% of overlap with initiation events detected by nanopore sequencing and 13.6% with PfORC binding sites [36]. Another group found no significant correlation between origins detected by DNAscent nanopore sequencing and $PfORC1_{1\text{-}2}$ binding sites within the same organism [63]. Furthermore, there is low overlap between origins mapped using different techniques and ORC-binding sites in human cells [59]. We did not detect any accumulation of origins detected by stranded SNS-seq within SSRs. This differs from the previously reported accumulation of TbORC1/CDC6 binding sites at divergent SSRs in the Tb927 genome [42]. The reason for the low correlation between origins detected by different methods and ORC binding sites remains unknown, and further studies are necessary to answer this complex question.

It is important to note that the topology of origins of *T. brucei* is very similar to that previously described for other eukaryotes. The origins in mouse cells have been characterised as fragile sites, defined by long poly (dA:dT) tracks that are nucleosome free and devoid of the single-strand DNA-protecting protein RPA [13]. The same position and orientation of predicted G4 motifs, peaking at 240-300 base pairs from replication start sites, was previously identified in two drosophila cell lines using conventional SNS-seq [21], as was found here for *T. brucei*. Furthermore, the organisation of nucleosomes around the origins of *T. brucei* is identical to that observed in yeast and higher eukaryotes [12,28–32,64,65]. In the case of *L. major* the origins identified using conventional SNS-seq were also located in genomic regions with lower nucleosome occupancy [22]. In addition, a significant correlation between *L. major* origins and predicted G4 motifs was observed, with 74% of origins containing at least one such motif [22]. These similarities provide strong support for the reliability of our origin identification in *T. brucei* and suggest that the structure of origins may be a conserved feature in all eukaryotes. However, further studies are needed to gain a deeper insight into the evolution of DNA replication origins.

Methods

Cell culture

Procyclic forms (PCF) of *T. brucei*, strain Lister 427 wild type strain were grown at 27 °C in SDM-79 (PAA Laboratories) supplemented with heat inactivated 10% FBS (Gibco) and 7 µg.mL^{-1} hemin [66]. Bloodstream forms (BSF) of *T. brucei* Lister 427 strain were cultivated at 37°C, 5% CO in a modified Iscove's culture medium (Gibco) complemented according to Hirumi *et al.* [67], with 10% (v/v) heat-inactivated foetal bovine serum (Gibco), 36 mM sodium bicarbonate (Sigma), 136 μg.mL^{-1} hypoxanthine (Sigma), 39 μg.mL^{-1} thymidine (Sigma), 110 μg.mL^{-1} Na-pyruvate (Sigma), 28 μg.mL^{-1} bathocuproine (Sigma), 0.25 mM β-mercaptoethanol, 2 mM L-cysteine.

SNS isolation and purification

SNS were isolated and purified following a previously published protocol [16] with few modifications. Between 2 - 2.5×10^9 cells were used as starting material for SNS isolation. Total DNA was isolated from three biological replicates of asynchronous PCF and BSF cells in the exponential phase of growth (3-5×10^6 and 7-9×10^5 cells.mL^{-1} for PCF and BSF respectively). Cells were precipitated and incubated in 0.5 mL of 20 mM Tris-Cl, pH 7.0, 50 mM EDTA, 1% N-Lauroylsarcosine, 0.5% Triton X-100 and 2 mg.mL^{-1} proteinase K and incubated at 45°C for 24 hours. Fresh solution of proteinase K was added after 24 hours and digested for 16 hours more. DNA was purified with phenol/chloroform/isoamyl alcohol and precipitated in 0.3 M Na-acetate (pH 5.3) and absolute ethanol. DNA was dissolved gently in TEN20 buffer (10 mM Tris-Cl, 50 mM EDTA, 20 mM NaCl, pH 7.0) with 100 U RNasin (Promega). DNA was heat-denatured (95°C/5 min) and loaded onto 5 mL of neutral 5%–20% sucrose gradients prepared in TEN20 buffer in ultra-clear tubes (Beckman Coulter). Size separation of nucleic acids was done by ultra-centrifugation in SW50 rotor for 7 hours at 45 krpm at +4°C. Fractions of 200 μL from the sucrose gradient were collected and checked by agarose gel electrophoresis to assess the quality of nucleic acid separation (Supplementary Figure 10A). Fractions from 0.5-2.5 ×10^3 nt were pooled together as SNS enriched fractions. Half of the SNS pool was treated with 10 μg of cocktail of RNase, DNase free (Roche, Cat Nb 11119915001) for 16h at 37°C, to become the SNS-depleted control. SNS-enriched samples, together with its SNS-depleted controls, were treated 3-4 times with 50 U of T4 PNK (Thermo Fisher Scientific) and 100 U of λ-exonuclease (Thermo Fisher Scientific). Each treatment was done after heat denaturation of DNA (95°C/10 min) and simultaneously on samples and controls. After denaturation, samples and controls were immediately placed on ice. The enzymatic reactions were assembled on ice, and then the enzymatic treatment with T4 PNK and λ-exonuclease was done at 37°C. After enzymatic treatment and prior to library preparation, residual RNA was removed with 5 μg RNase, DNase-free (Roche Cat Nb 11119915001), isolated DNA was sonicated with Bioruptor Pico sonication device (Diagenode) and purified with magnetic beads (CleanNGS, Proteigen). The quantity of purified material was estimated by fluorometer DS 11 FX (DeNovix) using Qubit ssDNA Assay kit (Invitrogen).

It is important to note that we used SNS enrichment conditions and enzymatic treatments as described in previous reports [16,68]. We enriched the SNS by size on a sucrose gradient and then treated this SNS-enriched fraction with high amounts of λ-exonuclease. A previous study has shown that under these conditions, complete digestion of DNA containing G4-rich sequences is achieved [20]. The efficiency of the T4 PNK and λ-exonuclease enzymes was tested on both sheared gDNA and on RNA. The majority of DNA was digested after one treatment with these two enzymes (Supplementary Figure 10B). RNA molecules treated under the same conditions were preserved (Supplementary Figure 10C), suggesting that SNS would also be preserved. The quantity of DNA molecules in SNS containing fractions was systematically followed by qPCR after each enzymatic treatment and it was estimated that the amount of DNA in starting material was reduced around 100 times after the first treatment, and around 10 times more after each subsequent treatment with these two enzymes.

Preparation of stranded libraries for paired end sequencing

Single-stranded SNS libraries were prepared using the Accel-NGS 1S Plus DNA Library Kit and 1S Plus Dual Indexing Kit for Illumina (Swift Biosciences), following the manufacturer's protocol. A 3'end specific tail is preserving the strand orientation of the ssDNA fragments during library preparation, resulting in a second strand library, where the plus-strand fragments are represented by F1R2 read pairs and the minus-strand fragment by F2R1 read pairs. The quality of the libraries was checked by capillary electrophoresis on Bioanalyzer 2100 (Agilent) using Agilent High Sensitivity DNA Kit (Agilent Technologies). The quantity of libraries was estimated by fluorometry using dsDNA High Sensitivity Assay Kit on a DS-11 FX+ (DeNovix).

Paired end sequencing (2×150)

The libraries were validated for quality and quantity using Fragment Analyzer and qPCR techniques. qPCR was performed using KAPA Library Quantification Kits - Complete kit (Universal) (Roche, ref.7960140001) and Light Cycler 480 machine (Roche). The libraries were pooled equimolarly and sequenced on the Illumina NovaSeq 6000 using a NovaSeq Reagent Kit for 300 cycles (Illumina) following manufacturer's instructions. Libraries were sequenced on one lane of a SP flow-cell in paired-end 150 nt. Sequencing was performed by the GenomiX (MGX) Core Facility, Montpellier, France. Demultiplexing and FASTQ files production was done using the Illumina software bcl2fastq (*RRID:SCR_015058*, v2.20.0.422).

SNS-seq data analysis

Quality control of the FASTQ files was performed using FastQC (*RRID:SCR_014583*, v0.11.9) (*https://www.bioinformatics.babraham.ac.uk/projects/fastqc/*). The sequencing reads were tail trimmed to remove the 3'end specific tail introduced during library preparation using cutadapt [69] with the following parameters (-u -10 -u 10 -U -10 -U 10). Afterwards the SNS-seq sequencing reads were aligned against the entire *T. brucei* Lister 427-2018 reference genome release 55 from TriTrypDB (*RRID:SCR_007043*) (*https://tritrypdb.org/common/downloads/release-55/TbruceiLister427_2018*, last accessed June 24th, 2025) with bowtie2 (*RRID:SCR_016368*, v2.4.10) [70]. The alignment output bam files were converted, sorted, and indexed using samtools [71]. Duplicated reads were removed using picard (*RRID:SCR_006525*, v2.23.5, "Picard Toolkit." 2019. Broad Institute, GitHub Repository. *https://broadinstitute.github.io/picard/*, last accessed June 24th, 2025; Broad Institute). Mapped deduplicated reads pairs were then separated with samtools (*RRID:SCR_002105*, v1.13) *view* into minus-strand and plus-strand read pairs via their read orientation (minus-strand: F2R1 and plus-strand: F1R2). Peak calling was then performed with macs2 (v.2.2.7.1) (using default parameters plus -p 5e-2 -s 130 -m 10 30 --gsize 2.5e7) for plus and minus-strand reads separately with the SDC as control.

Origin selection

Origins were selected after the peak calling step, during the peak filtering step. The called peaks were filtered using the bedtools suite (*RRID:SCR_006646*, v.2.30.0) [72] as described below. In this peak filtering step, three criteria were defined to select the origins. The first criterion was that two SNS peaks must be oriented in a way that reflects the topology of the SNS strands within an active origin: the upstream peak must be on the minus strand and followed by the downstream peak on the plus strand. The second criterion was to set a maximal inner to inner border distance between two divergent minus and plus peak pair. The distances between the start points of the two divergent leading strands are not known. We conducted tests on several inner border-to-border distances, ranging from 0.5 to 2.5 kb. Higher numbers of origins were found with larger distances. However, we finally applied a maximum distance of 0.5 kb, presuming that this corresponds more closely to the physiological distance between two divergent SNS molecules. The third criterion was introduced to prevent complete overlap of two divergent peaks, as this can only occur if the two SNSs elongate at the same rate in both directions, which is an unlikely scenario. Indeed, the synthesis of the Okazaki fragment and ligation to the 5' of the SNS requires several enzymatic processes, and we can assume that the elongation of SNS fibre is slower at the 5' end. Of note, complete overlap of minus and plus peaks was rarely observed (mean 6.5% (stdev 6.5%, min 0.8%, max 15.8%)). To define the maximum overlap between the minus and plus SNS peaks, a series of tests were conducted, varying the percentage of overlap from 40% to 90%. This resulted in a similar number of origins, and ultimately, 50% was selected as the optimal value. Therefore, firstly, minus-strand and plus-strand peaks with an overlap of over 50% were removed with bedtools *intersect*. Secondly peak pairs on different strands with a maximal distance of 0.5 kb were selected using bedtools *windows*. Finally, peak pairs were filtered for the correct orientation (minus-strand followed by plus-strand) with bedtools *closest* and the origin was defined as the genomic region between the two inner borders of two divergent peaks. The overlap of origins between replicates and cell types as well as other genomic features was analysed with bedtools *windows* and bedtools *intersect*. Replicates were merged with bedtools *merge*.

For **visual inspection of genomic features** Integrative genomics viewer (IGV) (*RRID:SCR_011793*) [73] was used. Violin plots were generated with the GraphPad Prism 8.3.0 software (*RRID:SCR_002798*, GraphPad Software Inc., La Jolla, CA, USA).

Shuffled controls were generated with bedtools *shuffle* (-chrom –noOverlapping –seed - excl) that randomly permutes genomic features provided respecting the number, the size and chromosomal location and either excluding the detected origins or excluding detected origins and coding sequences (called intergenic shuffled control).

Clustering of origins origin clustering was analysed with bedtools *cluster* using different distances.

The inter-origin distance (IOD) between the origins detected by SNS-seq was calculated using bedtools *closest* (-io (ignore overlapping) -iu (ignore upstream)) on a single file.

Read sub-sampling was performed to be able to compare origin number in BSF and PCF. The mapped reads from all three PCF samples and control and BSF samples and control, respectively, were merged using Samtools *merge*. The PCF combined sample was sub-samples with Samtools *view* (--subsample 0.69 --subsample-seed [0-14]) and the merged control with Samtools *view* (--subsample 0.57 --subsample-seed [0-14]) 15 times to obtain a set of mapped reads comparable to the BSF combined sample. Origin detection was applied as described above to the combined BSF and the 15 subsamples combined PCF samples.

For **visualization of the respective** localization of two or more datasets the the Deeptools suite (*RRID:SCR_016366*, v3.5.1) [74] and the ggplot2 R package was used. The open source ggplot2 R package (*RRID:SCR_014601*), freely available from the Comprehensive R Archive Network (CRAN) repository (*http://www.r-project.org*, last accessed June 24th, 2025) was used to generate composite plots. In Deeptools bed and bigwig files were used to generate a matrix file with *ComputeMatrix* and *PlotHeatmap* or *PlotProfil* were used to generate visualizations from the matrix.

Motive enrichments were tested over the origin centres extended 250 bp up and downstream with the MEME suite motive discovery (*RRID:SCR_001783*) [75] as well as with homer (*RRID:SCR_010881*) *findMotifs* [76]. A multiple sequence alignment and the analysis of nucleotide patterns were built with the WebLogo (*RRID:SCR_010236*) online application [47].

Nucleotide composition around the origin centres was calculated as follows. The *T brucei* genome was split into 20 bp windows using bedtools *window*. The number of A, T, G and C nucleotides for each window were obtained with bedtools *nuc* and used to calculate the nucleotide percentage per 20 bp window. The resulting bedgraph files were transformed into bigwig file format using USCS utilities *bedGraphToBigWig* (*https://github.com/ucscGenomeBrowser/kent/tree/master/src/utils*, last accessed June 24th, 2025).

Comparison of origins mapped by stranded SNS-seq with previously published datasets

Handling of datasets mapped to different reference genomes

The stranded SNS-seq data, that we mapped to the Tb Lister 427-2018 genome, were compared with various published datasets (SAS and PAS (TriTrypDB), MNase-seq [50], DRIP-seq [54], G4-seq [49], MFA-seq [42] and Orc1/CDC6 ChIP-on-Chip [42]) that were analysed using different references genomes (Tb Lister 427 or TREU927). To enable direct comparison, we either reanalysed the published datasets against the Tb Lister 427-2018 genome or remapped the stranded SNS-seq data to match the respective reference genomes of the compared datasets. The mapping of origins in the different reference genomes produced different numbers of origins, in Tb Lister 427 1059 PCF and

1466 BSF origins and in total 2081 origins were detected, in TREU927 976 PCF and 1466 BSF origins were detected and in total 1963 origins were detected. Specific details on the chosen approach are given in the relevant paragraphs below.

Splice leader acceptor sites (SAS) and polyadenylation sites (PAS)

were obtained from TriTrypDB. The tracks 'unified spliced leader addition sites' and the 'unified poly A sites' were downloaded from the genome browser view for the 11 Megabase chromosomes of the Tbb TREU927 reference genome in bed file format. We compared these datasets to SNS-seq dataset analysed as described above in the Tbb TREU927 reference genome (*https://tritrypdb.org/common/downloads/release-62/TbruceiTREU927/*).

Transcription start sites (TSS), transcription termination sites (TTS) and HT regions

A list of TSS, TTS and head-to-tail (HT) regions for the Tbb TREU927 reference genome was obtained from Kim et al. [56].

MNase-seq

The MNase-seq data [50] was downloaded as bigwig files from GEO (accession number GSE90593). The data was published using the Tb Lister 427 reference genome. To be able to compare our SNS-seq dataset with this data we reanalysed the SNS-seq sequencing reads as described above in the Tb Lister 427 reference genome (*https://tritrypdb.org/common/downloads/release-62/TbruceiLister427/* last accessed June 24th, 2025). To quantify the MNase-seq signature we observed around the origin centres ("M" pattern), we extracted the average MNase-seq signal (normalised dyad values [50]) from the regions 400-200 nt upstream, 200-400 nt downstream and 75nt up- and downstream of the origin centre with the UCSC utilities *bigWigAverageOverBed* command (*https://github.com/ucscGenomeBrowser/kent/tree/master/src/utils* last accessed June 24th, 2025). Regions with negative values representing tandem repeats and N-rich sequences [50] were discarded. LNO regions were considered as present when the average MNase-seq value was below 0.23 normalised dyad value and the HNO regions were considered present when the averaged normalised dyad value for the region was above this value. This value corresponds to the average value for background regions (2000-1000 bp upstream of the origin centres and 1000-2000 bp downstream of origin centres).

Drip-seq

The Drip-seq dataset was downloaded from ENA (accession number PRJEB21868) [54] as fastq files. The re-analysis was done as described [54]. The published DRIP-seq analysis was performed in Tb Lister 427 reference genome. We re-analysed the data in the Tb Lister 427 and the Tb Lister 427-2018 reference genomes. The intersection of DRIP-seq data with the stranded SNS-seq origins was done in the Tb Lister 427-2018 reference genome.

G4-seq

The G4-seq [49] dataset was downloaded from GEO (accession number GSE110582) as bed and bedgraph files separated for the plus and minus strand. G4-seq was performed on *T. brucei* EATR01125 strain and analysed in the Tbb TREU927 reference genome [49] to be able to directly compare the data with the SNS-seq dataset we analysed the SNS-deq data in the Tbb TREU927 reference genome.

G4 prediction

in the *T. brucei* genome were done with G4Hunter [51] with the following settings, window size (-w 25) and a threshold of 1.57. Prediction of G4 structures in the Tb 427-2018 reference genome and in the Tb Lister 427 reference genome were performed, to be able to compare G4 structures to MNase-seq (in Tb Lister 427 reference genome) and DRIP-seq (in Tb Lister 427-2018 reference genome) data as G4-seq output files were available only for the Tbb TREU927 reference genome [49].

MFA-seq

We intersected six MFA-seq datasets. Two were obtained directly from the authors [42] and four were downloaded from ENA (accession number PRJEB11437) [43], all in fastq file format. The analysis of the fastq files was performed as described previously [42,43]. The original published MFA-seq analysis was performed in *T. brucei* Tbb TREU927 reference genome. We re-analyzed the data in the Tbb TREU927 reference genome and in the Tb Lister 427-2018 reference genomes to be able to directly compare them to the stranded SNS-seq data.

ORC1/CDC6 ChIP-on-chip

The Orc1/CDC6 ChIP-on-Chip dataset [42] was downloaded from TriTrypDB (as ratios_peaks) for each element in the *T. brucei* Tbb TREU927 reference genome in form of bed files. The dataset available on TriTrypDB differs from the one published in the manuscript of Tiengwe et al. [42] but the originally published dataset is not available anymore (personal communication Richard McCulloch). The TriTrypDB ORC1/CDC6 dataset contains 6543 binding sites whereas the dataset published originally contained 953 binding sites [42], to obtain a number comparable to the published number of ORC1/CDC6 binding sites we filtered the TritrypBD data set at a score of 22 which resulted in a set of 990 ORC1/CDC6 binding sites. To be able to directly compare them to the SNS-seq we analysed the SNS-seq data in the Tbb TREU927 reference genome.

DNA molecular combing

Asynchronous cell populations of *T. brucei* were grown to the density of 4×10^6 and 7×10^5 cells.mL^{-1} for PCF and BSF, respectively. Cells were sequentially labelled with two modified nucleosides, firstly with 300 µM iodo-deoxyuridine (IdU, Sigma) for 20 minutes and then 20 minutes with 300 µM chloro-deoxyuridine (CldU, Sigma) without intermediate washing. After labelling, the cells were immediately placed on ice to stop DNA replication and extensively washed with ice-cold 1xPBS. Around 1×10^8 cells were blocked in 1% low melting agarose plugs.

DNA molecular combing was done as described previously [41]. Protein-free plugs with DNA were extensively washed in TEN20 buffer (10 mM Tris-Cl, 50 mM EDTA, 20 mM NaCl, pH 7.0) and then stained with 0.5 µM YOYO-1 fluorescent dye (Molecular Probes) for 1 h at 37oC. After washing from the excess YOYO-1 day, the 200 µL of TEN20 buffer was added to the plugs, melted at 65 °C for 15 minutes, slowly cooled down to 42 °C and treated with 10 U of β-agarase (Merck) for 16h. This treatment was repeated with fresh β-agarase for 4 hours. After digestion, 2 mL of 50 mM MES (mix of 0.5 M MES Hydrate and 0.5M MES Sodium Salt, pH 5.7) was added very gently to the DNA solution and then DNA molecules were combed and regularly stretched (2 kb/µm) on silanized coverslips provided by Montpellier DNA Combing Facility. Linearized DNA molecules were fixed for 16 hours at 65 °C, denatured in 1N NaOH for 20 minutes and blocked with 1 × PBS/ 1% BSA/ 0.1% Triton X-100. Immuno-detection was done with antibodies diluted in 1 × PBS/ 1% BSA/ 0.1% Triton X-100 and incubated at 37 °C in a humid chamber for 60 min. Each step of incubation with antibodies was followed by extensive washes with 1 × PBS. Immuno-detection was done with anti-ssDNA antibody (1/100 dilution, clone 16-19, Cat. No. MAB3034, Merck Milipore), the mouse anti-BrdU antibody (1/20 dilution, clone B44, Cat. No. 347580, Becton Dickinson) and rat anti-BrdU antibody (1/20 dilution, clone BU1/75, Cat. No. ABC117-7513, Abcys). The following secondary antibodies were used: Alexa488 Goat anti Rat IgG (1/50 dilution, Cat. No. A-11006, Invitrogen), Alexa 546 Goat anti Mouse IgG1 (1/50 dilution, Cat. No. A-21123, Invitrogen) and Alexa 647 goat anti-mouse IgG2a (1/100 dilution, Cat. No. A-21241, Invitrogen). Coverslips were mounted with 15 µL of Prolong Gold Antifade (Cat. No. P10144, Thermo Fisher Scientific) and dried at room temperature for 16 hrs. Images were acquired using Zeiss Z1 microscope, equipped with an ORCA-Flash 4.0 digital CMOS camera (Hamamatsu), using a 40× objective, where 1 pixel corresponds to 325 bp (one microscope field of view corresponds to ∼450 kb). Acquisition and analysis of images were done by MetaMorph version 7 software (Molecular Devices LLC) (*RRID:SCR_002368*).

Measurements and statistical analysis of combed DNA molecules

The speed of replication forks was estimated on individual forks, presented as IdU track followed by a CldU track. Only intact (non-fragmented) forks were measured, which was followed by DNA immuno-staining. The speed was calculated as the ratio between the CldU track length (kilobases) and the CldU pulse duration (minutes). The IODs were measured as the distance (in kilobases) between the centres of two adjacent progressing forks located on an uninterrupted DNA fibre. Fork asymmetry was calculated as the ratio of the longer track over the shorter track in progressing divergent forks. A ratio greater than one indicates asymmetry. The GraphPad Prism 8.3.0 software (GraphPad Software Inc., La Jolla, CA, USA) was used to generate graphs and to perform statistical analysis of DNA combing measurements as described previously [41]. The statistical comparisons of different datasets were assessed by the non-parametric Mann–Whitney two-tailed tests.

Data availability

The SNS-seq data (FASTQ files) and the BED files containing the positions of the origins of the merged and separated PCF and BSF samples from this study have been submitted to the European Nucleotide Archive under the accession number PRJEB70708. The SNS-seq data analysis pipeline is available at *https://github.com/bridlin/finding_ORIs* and *https://zenodo.org/records/14289282*.

Acknowledgements

Special thanks to Cedric Notredame for advising us to prepare SNS stranded libraries. We would like to thank Julie A.J. Clement and Stephanie Le Gras for their help with preliminary analysis of SNS-seq data. We acknowledge the Montpellier DNA Combing Facility for providing silanized coverslips. We are grateful to MGX-Montpellier GenomiX (Montpellier, France) facility for performing sequencing. We would also like to acknowledge the 'Eukaryotic Pathogen, Vector and Host Informatics Resource' (VEuPathDB) for the integrated database, covering numerous eukaryotic pathogens [77]. We are grateful to the Roscoff Bioinformatics platform ABiMS (*http://abims.sb-roscoff.fr*), part of the Institut Français de Bioinformatique (ANR-11-INBS-0013) and BioGenouest network, for providing computing resources. Thanks to Frédéric Bringaud (University Victor Segalen Bordeaux 2) for providing the T. brucei 427 wild-type PCF strain and Lucy Glover (Institute Pasteur, Paris) for providing us T. brucei 427 wildtype BSF strain.

The work was supported by the Agence Nationale de la Recherche (ANR) within the frame of the "Investissements d'avenir" program [ANR-11-LABX-0024-01 "PARAFRAP" to YS], the Centre National de la Recherche Scientifique (CNRS), the French Ministry of Research and the Centre Hospitalier Universitaire of Montpellier. SS was recipient of a grant from the Fondation pour la Recherche Médicale (ING20160435527). MGX acknowledges financial support from France Génomique National infrastructure, funded as part of "Investissement d'Avenir" program managed by Agence Nationale pour la Recherche (contract ANR-10-INBS-09). ABiMS is founded as a part of the Institut Français de Bioinformatique (ANR-11-INBS-0013).

Additional files

Supplementary data. Supplementary Tables 1-2 and Suppl Figures 1-7.

Additional information

Funding

Funder	Grant reference number
Agence Nationale de la Recherche	ANR-11-LABX-0024-01 "PARAFRAP"
Agence Nationale de la Recherche	ANR-10-INBS-09
Agence Nationale de la Recherche	ANR-11-INBS-0013

Fondation pour la Recherche Médicale	ING20160435527

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Peer reviews

Reviewer #1 (Public review):

In this paper, Stanojcic and colleagues attempt to map sites of DNA replication initiation in the genome of the African trypanosome, Trypanosoma brucei. Their approach to this mapping is to isolate 'short-nascent strands' (SNSs), a strategy adopted previously in other eukaryotes (including in the related parasite Leishmania major), which involves isolation of DNA molecules whose termini contain replication-priming RNA. By mapping the isolated and sequenced SNSs to the genome (SNS-seq), the authors suggest that they have identified origins, which they localise to intergenic (strictly, inter-CDS) regions within polycistronic transcription units and suggest display very extensive overlap with previously mapped R-loops in the same loci. Finally, having defined locations of SNS-seq mapping, they suggest they have identified G4 and nucleosome features of origins, again using previously generated data. Though there is merit in applying a new approach to understand DNA replication initiation in T. brucei, where previous work has used MFA-seq and ChIP of a subunit of the Origin Replication Complex (ORC), there are two significant deficiencies in the study that must be addressed to ensure rigour and accuracy.

(i) The suggestion that the SNS-seq data is mapping DNA replication origins that are present in inter-CDS regions of the polycistronic transcription units of T. brucei is novel and does not agree with existing data on the localisation of ORC1/CDC6, and it is very unclear if it agrees with previous mapping of DNA replication by MFA-seq due to the way the authors have presented this correlation. For these reasons, the findings essentially rely on a single experimental approach, which must be further tested to ensure SNS-seq is truly detecting origins. Indeed, in this regard, the very extensive overlap of SNS-seq signal with RNA-DNA hybrids should be tested further to rule out the possibility that the approach is mapping these structures and not origins.

(ii) The authors' presentation of their SNS-seq data is too limited and therefore potentially provides a misleading view of DNA replication in the genome of T. brucei. The work is presented through a narrow focus on SNS-seq signal in the inter-CDS regions within polycistronic transcription units, which constitute only part of the genome, ignoring both the transcription start and stop sites at the ends of the units and the large subtelomeres, which are mainly transcriptionally silent. The authors must present a fuller and more balanced view of SNS-seq mapping, across the whole genome, to ensure full understanding and clarity.

In the revised manuscript, the authors have improved the presentation and analysis of their data, expanding the description of SNS-seq mapping across the genome, and more clearly assessing to what extent there is correlation between SNS-seq signal and previous mapping approaches to predict origins (by MFA-seq and ChiP-chip of ORC1/CDC6). With regard the correlation between SNS-seq and ORC/1CDC6 ChIP-chip, it should be noted that two datasets were generated in distinct strains of T. brucei (Lister 427 and TREU927, respectively), and it is unclear if the latter dataset can be accurately mapped to the strain used here. Notwithstanding this concern, these improvements clarify a number of aspects of the SNS-seq mapping: (1) the signal is more prevalent in the transcribed core of the genome than in the largely transcriptionally silent subtelomeres; and (2) whereas previous work revealed strong correlation between ORC1/CDC6 localisation and MFA-seq peaks at the ends of multigene transcription units, neither of these data show significant overlap with SNS-seq signal, which is not seen at transcription start or stop sites ('SSRs'; supplementary Fig.8D) and shows marked depletion at predicted ORC1/CDC6 sites (supplementary Fig.8C). To the authors' credit, they acknowledge this lack of correlation in the discussion.

The authors have not provided any new data to substantiate their assertion that SNS-seq accurately detects origins in T. brucei, and therefore the work rests on a single experimental approach, without validation. As a result, the suggestion of abundant, previously undetected origins in the intergenic regions of multigene transcription remains a prediction. One key untested limitation of the work lies in the observation that the very large majority of SNS-seq signal overlaps with previously RNA-DNA hybrids; without an experimental test, the suggestion that the authors have 'disclosed for the first time a strong link between RNA:DNA hybrid formation and DNA replication initiation' remains conjecture.

https://doi.org/10.7554/eLife.108143.2.sa2

Reviewer #2 (Public review):

Summary:

Stanojcic et al. investigate the origins of DNA replication in the unicellular parasite Trypanosoma brucei. They perform two experiments, stranded SNS-seq and DNA molecular combing. Further, they integrate various publicly available datasets, such as G4-seq and DRIP-seq, into their extensive analysis. Using this data, they elucidate the structure of origins of replications. In particular, they find various properties located at or around origins, such as polynucleotide stretches, G-quadruplex structures, regions of low and high nucleosome occupancy, R-loops, and that origins are mostly present in intergenic regions. Combining their population-level SNS-seq and their single-molecule DNA molecular combing data, they elucidate the total number of origins as well as the number of origins active in a single cell.

Between the initial submission and this revision, the raised major concerns have not been resolved, and no additional validation has been provided.

Strengths:

(1) A very strong part of this manuscript is that the authors integrate several other datasets and investigate a large number of properties around origins of replication. Data analysis clearly shows the enrichment of various properties at the origins, and the manuscript is concluded with a very well-presented model that clearly explains the authors' understanding and interpretation of the data.

(2) The DNA combing experiment is an excellent orthogonal approach to the SNS-seq data. The authors used the different properties of the two experiments (one giving location information, one giving single-molecule information) well to extract information and contrast the experiments.

(3) The discussion is exemplary, as the authors openly discuss the strengths and weaknesses of the approaches used. Further, the discussion serves its purpose of putting the results in both an evolutionary and a trypanosome-focused context.

Weaknesses:

I have major concerns about the origin of replication sites determined from the SNS-seq data. As a caveat, I want to state that, before reading this manuscript, SNS-seq was unknown to me; hence, some of my concerns might be misplaced.

(1) There are substantial discrepancies between the origins identified here and those reported in previous studies. Given that the other studies precede this manuscript, it is the authors' duty to investigate these differences. A conclusion should be reached on why the results are different, e.g., by orthogonally validating origins absent in the previous studies.

(2) I am concerned that up to 96% percent of all SNS-seq peaks are filtered away. If there is so much noise in the data, how can one be sure that the peaks that remain are real? Upon request, the authors have performed a control, where randomly placed peaks were run through the same filtering process. Only approximately twice as many experimental peaks passed filtering compared to random peaks. While the authors emphasize reproducibility between replicates, technical artifacts from the protocol would also be reproducible. Moreover, in other SNS-seq studies, for example, Pratto et al. Cell 2021, Fig. 1B, + and − strand peaks always appear closely paired. This pattern contrasts strongly with Fig. 2A in this manuscript.

Further, I have some minor concerns that do not affect the main conclusions of the manuscript:

- Fig 2C: The regions shown in the heatmap have different sizes, and I presume that the regions are ordered by size on the y-axis? If so, does the cone-shaped pattern, which is origin-less for genic regions and origin-enriched for intergenic regions, arise from the size of the regions? (I.e., for each genic region, the region itself is origin-less and the flanking intergenic regions contain origins.) If this is the case, then the peaks/valleys, centered exactly on the center of the regions on the mean frequency plots, arise from the different sizes of the analyzed regions, not from the fact that origins are mostly found at the center of intergenic regions. This data would be better presented with all regions stretched to the same size. This has not been addressed in the revision.

- Line 123, "and the average length of origins was found to be approximately 150 bp.": To determine origins, the authors filter away overlapping peaks and peaks that are too far from each other. Both restrict the minimal and maximal length of origins that can be observed, and this, in turn, affects the average length. This has not been addressed in the revision.

Are claims well substantiated?:

The identification of origins via SNS-seq appears to be incompletely supported to me.

All downstream analyses depend on the reliability of origin identification.

Impact:

This study has the potential to be valuable for two fields: In research focused on T. brucei as a disease agent, where essential processes that function differently than in mammals are excellent drug targets. Further, this study would impact basic research analyzing DNA replication over the evolutionary tree, where T. brucei can be used as an early-divergent eucaryotic model organism.

https://doi.org/10.7554/eLife.108143.2.sa1

Author response:

The following is the authors' response to the current reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *In this paper, Stanojcic and colleagues attempt to map sites of DNA replication initiation in the genome of the African trypanosome, Trypanosoma brucei. Their approach to this mapping is to isolate 'short-nascent strands' (SNSs), a strategy adopted previously in other eukaryotes (including in the related parasite Leishmania major), which involves isolation of DNA molecules whose termini contain replication-priming RNA. By mapping the isolated and sequenced SNSs to the genome (SNS-seq), the authors suggest that they have identified origins, which they localise to intergenic (strictly, inter-CDS) regions within polycistronic transcription units and suggest display very extensive overlap with previously mapped R-loops in the same loci. Finally, having defined locations of SNS-seq mapping, they suggest they have identified G4 and nucleosome features of origins, again using previously generated data. Though there is merit in applying a new approach to understand DNA replication initiation in T. brucei, where previous work has used MFA-seq and ChIP of a subunit of the Origin Replication Complex (ORC), there are two significant deficiencies in the study that must be addressed to ensure rigour and accuracy.*
>
> *(i) The suggestion that the SNS-seq data is mapping DNA replication origins that are present in inter-CDS regions of the polycistronic transcription units of T. brucei is novel and does not agree with existing data on the localisation of ORC1/CDC6, and it is very unclear if it agrees with previous mapping of DNA replication by MFA-seq due to the way the authors have presented this correlation. For these reasons, the findings essentially rely on a single experimental approach, which must be further tested to ensure SNS-seq is truly detecting origins. Indeed, in this regard, the very extensive overlap of SNS-seq signal with RNA-DNA hybrids should be tested further to rule out the possibility that the approach is mapping these structures and not origins.*
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> *(ii) The authors' presentation of their SNS-seq data is too limited and therefore potentially provides a misleading view of DNA replication in the genome of T. brucei. The work is presented through a narrow focus on SNS-seq signal in the inter-CDS regions within polycistronic transcription units, which constitute only part of the genome, ignoring both the transcription start and stop sites at the ends of the units and the large subtelomeres, which are mainly transcriptionally silent. The authors must present a fuller and more balanced view of SNS-seq mapping, across the whole genome, to ensure full understanding and clarity.*
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supplementary Fig.8D) and shows marked depletion at predicted ORC1/CDC6 sites (supplementary Fig.8C). To the authors' credit, they acknowledge this lack of correlation in the discussion.

The authors have not provided any new data to substantiate their assertion that SNS-seq accurately detects origins in T. brucei, and therefore the work rests on a single experimental approach, without validation. As a result, the suggestion of abundant, previously undetected origins in the intergenic regions of multigene transcription remains a prediction. One key untested limitation of the work lies in the observation that the very large majority of SNS-seq signal overlaps with previously RNA-DNA hybrids; without an experimental test, the suggestion that the authors have 'disclosed for the first time a strong link between RNANA hybrid formation and DNA replication initiation' remains conjecture.

Reviewer #2 (Public review):

Summary:

Stanojcic et al. investigate the origins of DNA replication in the unicellular parasite Trypanosoma brucei. They perform two experiments, stranded SNS-seq and DNA molecular combing. Further, they integrate various publicly available datasets, such as G4-seq and DRIP-seq, into their extensive analysis. Using this data, they elucidate the structure of origins of replications. In particular, they find various properties located at or around origins, such as polynucleotide stretches, G-quadruplex structures, regions of low and high nucleosome occupancy, R-loops, and that origins are mostly present in intergenic regions. Combining their population-level SNS-seq and their single-molecule DNA molecular combing data, they elucidate the total number of origins as well as the number of origins active in a single cell.

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(3) The discussion is exemplary, as the authors openly discuss the strengths and weaknesses of the approaches used. Further, the discussion serves its purpose of putting the results in both an evolutionary and a trypanosome-focused context.

Weaknesses:

I have major concerns about the origin of replication sites determined from the SNS-seq data. As a caveat, I want to state that, before reading this manuscript, SNS-seq was unknown to me; hence, some of my concerns might be misplaced.

(1) There are substantial discrepancies between the origins identified here and those reported in previous studies. Given that the other studies precede this manuscript, it is the authors' duty to investigate these differences. A conclusion should be reached on why

> *the results are different, e.g., by orthogonally validating origins absent in the previous studies.*

We agree that orthogonally validation of origins detected by stranded SNS-seq is necessary and we are working on it.

> *(2) I am concerned that up to 96% percent of all SNS-seq peaks are filtered away. If there is so much noise in the data, how can one be sure that the peaks that remain are real? Upon request, the authors have performed a control, where randomly placed peaks were run through the same filtering process. Only approximately twice as many experimental peaks passed filtering compared to random peaks. While the authors emphasize reproducibility between replicates, technical artifacts from the protocol would also be reproducible. Moreover, in other SNS-seq studies, for example, Pratto et al. Cell 2021, Fig. 1B, + and − strand peaks always appear closely paired. This pattern contrasts strongly with Fig. 2A in this manuscript.*

The size and overlap of peaks depend on the length of the SNS. In our study, the width of the peaks corresponds to the size of the short nascent strands (0.5–2.5 kb) chosen as the starting material, whereas the width of the peaks in Pratto et al., Cell, 2021 are much larger (few kb). This could be due to the longer SNS used in the Pratto et al. study. Consequently, the overlap of the longer SNS is more pronounced since the SNS fibres elongate in both directions: at the 3' end by DNA polymerase and at the 5' end by ligation of Okazaki fragments. Additionally, the genomic regions displayed in our Figure 2A and in Pratto et al, Figure 1B are presented at substantially different resolutions, with a roughly ten-fold difference in scale.

> *Further, I have some minor concerns that do not affect the main conclusions of the manuscript:*
>
> *- Fig 2C: The regions shown in the heatmap have different sizes, and I presume that the regions are ordered by size on the y-axis? If so, does the cone-shaped pattern, which is origin-less for genic regions and origin-enriched for intergenic regions, arise from the size of the regions? (I.e., for each genic region, the region itself is origin-less and the flanking intergenic regions contain origins.) If this is the case, then the peaks/valleys, centered exactly on the center of the regions on the mean frequency plots, arise from the different sizes of the analyzed regions, not from the fact that origins are mostly found at the center of intergenic regions. This data would be better presented with all regions stretched to the same size. This has not been addressed in the revision.*

As the reviewer suggested, we have produced scaled plots of the stranded SNS-seq origins over genic and intergenic regions (see Figure 3, which is attached along with the Reviewer #2 (Recommendations for the authors)). However, we would prefer to keep the unscaled versions in the manuscript and add a note in the text as part of the Version of Record, explaining that the origins are evenly distributed throughout intergenic regions rather than being centred within them.

> *- Line 123, "and the average length of origins was found to be approximately 150 bp.": To determine origins, the authors filter away overlapping peaks and peaks that are too far from each other. Both restrict the minimal and maximal length of origins that can be observed, and this, in turn, affects the average length. This has not been addressed in the revision.*

This observation is correct. By applying filtering and setting the maximum distance between the positive and negative peaks, we are most likely affecting the average length by excluding potentially wider origins.

We'll modify the text as part of the Version of Record.

> *Are claims well substantiated?:*
>
> *The identification of origins via SNS-seq appears to be incompletely supported to me.*
>
> *All downstream analyses depend on the reliability of origin identification.*
>
> *Impact:*
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> *This study has the potential to be valuable for two fields: In research focused on T. brucei as a disease agent, where essential processes that function differently than in mammals are excellent drug targets. Further, this study would impact basic research analyzing DNA replication over the evolutionary tree, where T. brucei can be used as an early-divergent eucaryotic model organism.*

The following is the authors' response to the original reviews.

> ***eLife Assessment***
>
> *The authors use sequencing of nascent DNA (DNA linked to an RNA primer, "SNS-Seq") to localise DNA replication origins in Trypanosoma brucei, so this work will be of interest to those studying either Kinetoplastids or DNA replication. The paper presents the SNS-seq results for only part of the genome, and there are significant discrepancies between the SNS-Seq results and those from other, previously-published results obtained using other origin mapping methods. The reasons for the differences are unknown and from the data available, it is not possible to assess which origin-mapping method is most suitable for origin mapping in T. brucei. Thus at present, the evidence that origins are distributed as the authors claim - and not where previously mapped - is inadequate.*

We would like to clarify a few points regarding our study. Our primary objective was to characterise the topology and genome-wide distribution of short nascent-strand (SNS) enrichments. The stranded SNS-seq approach provides the high strand-specific resolution required to analyse origins. The observation that SNS-seq peaks (potential origins) are most frequently found in intergenic regions is not an artefact of analysing only part of the genome; rather, it is a result of analysing the entire genome.

We agree that orthogonal validation is necessary. However, neither MFA-seq nor TbORC1/CDC6 ChIP-on-chip has yet been experimentally validated as definitive markers of origin activity in T. brucei, nor do they validate each other.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *In this paper, Stanojcic and colleagues attempt to map sites of DNA replication initiation in the genome of the African trypanosome, Trypanosoma brucei. Their approach to this mapping is to isolate 'short-nascent strands' (SNSs), a strategy adopted previously in other eukaryotes (including in the related parasite Leishmania major), which involves isolation of DNA molecules whose termini contain replication-priming RNA. By mapping the isolated and sequenced SNSs to the genome (SNS-seq), the authors suggest that they have identified origins, which they localise to intergenic (strictly, inter-CDS) regions within polycistronic transcription units and suggest display very extensive overlap with previously mapped R-loops in the same loci. Finally, having defined locations of SNS-seq mapping, they suggest they have identified G4 and nucleosome features of origins, again using previously generated data.*
>
> *Though there is merit in applying a new approach to understand DNA replication initiation in T. brucei, where previous work has used MFA-seq and ChIP of a subunit of*

the Origin Replication Complex (ORC), there are two significant deficiencies in the study that must be addressed to ensure rigour and accuracy.

(1) The suggestion that the SNS-seq data is mapping DNA replication origins that are present in inter-CDS regions of the polycistronic transcription units of T. brucei is novel and does not agree with existing data on the localisation of ORC1/CDC6, and it is very unclear if it agrees with previous mapping of DNA replication by MFA-seq due to the way the authors have presented this correlation. For these reasons, the findings essentially rely on a single experimental approach, which must be further tested to ensure SNS-seq is truly detecting origins. Indeed, in this regard, the very extensive overlap of SNS-seq signal with RNA-DNA hybrids should be tested further to rule out the possibility that the approach is mapping these structures and not origins.

(2) The authors' presentation of their SNS-seq data is too limited and therefore potentially provides a misleading view of DNA replication in the genome of T. brucei. The work is presented through a narrow focus on SNS-seq signal in the inter-CDS regions within polycistronic transcription units, which constitute only part of the genome, ignoring both the transcription start and stop sites at the ends of the units and the large subtelomeres, which are mainly transcriptionally silent. The authors must present a fuller and more balanced view of SNS-seq mapping across the whole genome to ensure full understanding and clarity.

Regarding comparisons with previous work:

- Two other attempts to identify origins in *T. brucei* - ORC1/CDC6 binding sites (ChIP-on-chip, PMID: 22840408) and MFA-seq (PMID: 22840408, 27228154) - were both produced by the McCulloch group. These methods do not validate each other; in fact, MFA-seq origins overlap with only 4.4% of the 953 ORC1/CDC6 sites (PMID: 29491738). Therefore, low overlap between SNS-seq peaks and ORC1/CDC6 sites cannot disqualify our findings. Similar low overlaps are observed in other parasites (PMID: 38441981, PMID: 38038269, PMID: 36808528) and in human cells (PMID: 38567819).

- We also would like to emphasize that the ORC1/CDC6 dataset originally published (PMID: 22840408) is no longer available; only a re-analysis by TritrypDB exists, which differs significantly from the published version (personal communication from Richard McCulloch). While the McCulloch group reported a predominant localization of ORC1/CDC6 sites within SSRs at transcription start and termination regions, our re-analysis indicates that only 10.3% of TbORC1/CDC6-12Myc sites overlapped with 41.8% of SSRs.

- MFA-seq does not map individual origins, it rather detects replicated genomic regions by comparing DNA copy number between S- and G1-phases of the cell cycle (PMID: 36640769; PMID: 37469113; PMID: 36455525). The broad replicated regions (0.1–0.5 Mbp) identified by MFA-seq in *T. brucei* are likely to contain multiple origins, rather than just one. In that sense we disagree with the McCulloch's group who claimed that there is a single origin per broad peak. Our analysis shows that up to 50% of the origins detected by stranded SNS-seq locate within broad MFA-seq regions. The methodology used by McCulloch's group to infer single origins from MFA-seq regions has not been published or made available, as well as the precise position of these regions, making direct comparison difficult.

Finally, the genomic features we describe—poly(dA/dT) stretches, G4 structures and nucleosome occupancy patterns—are consistent with origin topology described in other organisms.

On the concern that SNS-seq may map RNA-DNA hybrids rather than replication origins: Isolation and sequencing of short nascent strands (SNS) is a well-established and widely used technique for high-resolution origin mapping. This technique has been employed for decades in various laboratories, with numerous publications documenting its use. We followed the published protocol for SNS isolation (Cayrou et al., Methods, 2012, PMID: 22796403). RNA-DNA hybrids cannot persist through the multiple denaturation steps in our workflow, as they melt at 95°C (Roberts and Crothers, *Science*, 1992; PMID: 1279808). Even in the unlikely event that some hybrids remained, they would not be incorporated into libraries prepared using a single-stranded DNA protocol and therefore would not be sequenced (see Figure 1B and Methods).

Furthermore, our analysis shows that only a small proportion (1.7%) of previously reported RNA-DNA hybrids overlap with SNS-seq origins. It is important to note that RNA-primed nascent strands naturally form RNA-DNA hybrids during replication initiation, meaning the enrichment of RNA-DNA hybrids near origins is both expected and biologically relevant.

On the claim that our analysis focuses narrowly on inter-CDS regions and ignores other genomic compartments: this is incorrect. We mapped and analyzed stranded SNS-seq data across the entire genome of *T. brucei* 427 wild-type strain (Müller et al., Nature, 2018; PMID: 30333624), including both core and subtelomeric regions. Our findings indicate that most origins are located in intergenic regions, but all analyses were performed using the full set of detected origins, regardless of location.

We did not ignore transcription start and stop sites (TSS/TTS). The manuscript already includes origin distribution across genomic compartments as defined by TriTrypDB (Fig. 2C) and addresses overlap with TSS, TTS and HT in the section "Spatial coordination between the activity of the origin and transcription". While this overlap is minimal, we have included metaplots in the revised manuscript for clarity.

> ***Reviewer #2 (Public review):***
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> *Summary:*
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> *Stanojcic et al. investigate the origins of DNA replication in the unicellular parasite Trypanosoma brucei. They perform two experiments, stranded SNS-seq and DNA molecular combing. Further, they integrate various publicly available datasets, such as G4-seq and DRIP-seq, into their extensive analysis. Using this data, they elucidate the structure of the origins of replication. In particular, they find various properties located at or around origins, such as polynucleotide stretches, G-quadruplex structures, regions of low and high nucleosome occupancy, R-loops, and that origins are mostly present in intergenic regions. Combining their population-level SNS-seq and their single-molecule DNA molecular combing data, they elucidate the total number of origins as well as the number of origins active in a single cell.*
>
> *Strengths:*
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> *(1) A very strong part of this manuscript is that the authors integrate several other datasets and investigate a large number of properties around origins of replication. Data analysis clearly shows the enrichment of various properties at the origins, and the manuscript concludes with a very well-presented model that clearly explains the authors' understanding and interpretation of the data.*

We sincerely thank you for this positive feedback.

> *(2) The DNA combing experiment is an excellent orthogonal approach to the SNS-seq data. The authors used the different properties of the two experiments (one giving location information, one giving single-molecule information) well to extract information and contrast the experiments.*

Thank you very much for this remark.

> *(3) The discussion is exemplary, as the authors openly discuss the strengths and weaknesses of the approaches used. Further, the discussion serves its purpose of putting the results in both an evolutionary and a trypanosome-focused context.*

Thank you for appreciating our discussion.

> *Weaknesses:*
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> *I have major concerns about the origin of replication sites determined from the SNS-seq data. As a caveat, I want to state that, before reading this manuscript, SNS-seq was unknown to me; hence, some of my concerns might be misplaced.*
>
> *(1) I do not understand why SNS-seq would create peaks. Replication should originate in one locus, then move outward in both directions until the replication fork moving outward from another origin is encountered. Hence, in an asynchronous population average measurement, I would expect SNS data to be broad regions of + and -, which, taken together, cover the whole genome. Why are there so many regions not covered at all by reads, and why are there such narrow peaks?*

Thank you for asking these questions. As you correctly point out, replication forks progress in both directions from their origins and ultimately converge at termination sites. However, the SNS-seq method specifically isolates short nascent strands (SNSs) of 0.5–2.5 kb using a sucrose gradient. These short fragments are generated immediately after origin firing and mark the sites of replication initiation, rather than the entire replicated regions. Consequently: (i) SNS-seq does not capture long replication forks or termination regions, only the immediate vicinity of origins. (ii) The narrow peaks indicate the size of selected SNSs (0.5–2.5 kb) and the fact that many cells initiate replication at the same genomic sites, leading to localized enrichment. (iii) Regions without coverage refer to genomic areas that do not serve as efficient origins in the analyzed cell population. Thus, SNS-seq is designed to map origin positions, but not the entire replicated regions.

> *(2) I am concerned that up to 96% percent of all peaks are filtered away. If there is so much noise in the data, how can one be sure that the peaks that remain are real? Specifically, if the authors placed the same number of peaks as was measured randomly in intergenic regions, would 4% of these peaks pass the filtering process by chance?*

Maintaining the strandness of the sequenced DNA fibres enabled us to filter the peaks, thereby increasing the probability that the filtered peak pairs corresponded to origins. Two SNS peaks must be oriented in a way that reflects the topology of the SNS strands within an active origin: the upstream peak must be on the minus strand and followed by the downstream peak on the plus strand.

As suggested by the reviewer, we tested whether randomly placed plus and minus peaks could reproduce the number of filter-passing peaks using the same bioinformatics workflow. Only 1–6% of random peaks passed the filters, compared with 4–12% in our experimental data, resulting in about 50% fewer selected regions (origins). Moreover, the “origins” from random peaks showed 0% reproducibility across replicates, whereas the experimental data showed 7–64% reproducibility. These results indicate that the retainee peaks are highly unlikely to arise by chance and support the specificity of our approach. Thank you for this suggestion.

> *(3) There are 3 previous studies that map origins of replication in T. brucei. Devlin et al. 2016, Tiengwe et al. 2012, and Krasiļņikova et al. 2025 (https://doi.org/10.1038/s41467-025-56087-3), all with a different technique: MFA-seq. All three previous studies mostly agree on the locations and number of origins. The authors compared their results to the first two, but not the last study; they found that their results are vastly different from the*

> *previous studies (see Supplementary Figure 8A). In their discussion, the authors defend this discrepancy mostly by stating that the discrepancy between these methods has been observed in other organisms. I believe that, given the situation that the other studies precede this manuscript, it is the authors' duty to investigate the differences more than by merely pointing to other organisms. A conclusion should be reached on why the results are different, e.g., by orthogonally validating origins absent in the previous studies.*

The MFA-seq data for *T. brucei* were published in two studies by McCulloch's group: Tiengwe et al. (2012) using TREU927 PCF cells, and Devlin et al. (2016) using PCF and BSF Lister427 cells. In Krasilnikova et al. (2025), previously published MFA-seq data from Devlin et al. were remapped to a new genome assembly without generating new MFA-seq data, which explains why we did not include that comparison.

Clarifying the differences between MFA-seq and our stranded SNS-seq data is essential. MFA-seq and SNS-seq interrogate different aspects of replication. SNS-seq is a widely used, high-resolution method for mapping individual replication origins, whereas MFA-seq detects replicated regions by comparing DNA copy number between S and G1 phases. MFA-seq identified broad replicated regions (0.1–0.5 Mb) that were interpreted by McCulloch's group as containing a single origin. We disagree with this interpretation and consider that there are multiple origins in each broad peaks; theoretical considerations of replication timing indicate that far more origins are required for complete genome duplication during the short S-phase. Once this assumption is reconsidered, MFA-seq and SNS-seq results become complementary: MFA-seq identifies replicated regions, while SNS-seq pinpoints individual origins within those regions. Our analysis revealed that up to 50% of the origins detected by stranded SNS-seq were located within the broad MFA peaks. This pattern—broad MFA-seq regions containing multiple initiation sites—has also recently been found in *Leishmania* by McCulloch's team using nanopore sequencing (PMID: 26481451). Nanopore sequencing showed numerous initiation sites within MFA-seq regions and additional numerous sites outside these regions in asynchronous cells, consistent with what we observed using stranded SNS-seq in *T. brucei*. We will expand our discussion and conclude that the discrepancy arises from methodological differences and interpretation. The two approaches provide complementary insights into replication dynamics, rather than 'vastly different' results.

We recognize the importance of validating our results in future using an alternative mapping method and functional assays. However, it is important to emphasize that stranded SNS-seq is an origin mapping technique with a very high level of resolution. This technique can detect regions between two divergent SNS peaks, which should represent regions of DNA replication initiation. At present, no alternative technique has been developed that can match this level of resolution.

> *(4) Some patterns that were identified to be associated with origins of replication, such as G-quadruplexes and nucleosomes phasing, are known to be biases of SNS-seq (see Foulk et al. Characterizing and controlling intrinsic biases of lambda exonuclease in nascent strand sequencing reveals phasing between nucleosomes and G-quadruplex motifs around a subset of human replication origins. Genome Res. 2015;25(5):725-735. doi:10.1101/gr.183848.114).*

It is important to note that the conditions used in our study differ significantly from those applied in the Foulk et al. Genome Res. 2015. We used SNS isolation and enzymatic treatments as described in previous reports (Cayrou, C. et al. Genome Res, 2015 and Cayrou, C et al. Methods, 2012). Here, we enriched the SNS by size on a sucrose gradient and then treated this SNS-enriched fraction with high amounts of repeated λ-exonuclease treatments (100u for 16h at 37oC - see Methods). In contrast, Foulk et al. used sonicated total genomic DNA for origin mapping, without enrichment of SNS on a sucrose gradient as we did, and then they performed a λ-exonuclease treatment. A previous study (Cayrou, C. et al. Genome Res, 2015,

Figure S2, which can be found at *https://genome.cshlp.org/content/25/12/1873/suppl/DC1*) has shown that complete digestion of G4-rich DNA sequences is achieved under the conditions we used.

Furthermore, the SNS depleted control (without RNA) was included in our experimental approach. This control represents all molecules that are difficult to digest with lambda exonuclease, including G4 structures. Peak calling was performed against this background control, with the aim of removing false positive peaks resulting from undigested DNA structures. We explained better this step in the revised manuscript.

The key benefit of our study is that the orientation of the enrichments (peaks) remains consistent throughout the sequencing process. We identified an enrichment of two divergent strands synthesised on complementary strands containing G4s. These two divergent strands themselves do not, however, contain G4s (see Fig. 8 for the model). Therefore, the enriched molecules detected in our study do not contain G4s. They are complementary to the strands enriched with G4s. This means that the observed enrichment of

G4s cannot be an artefact of the enzymatic treatments used in this study. We added this part in the discussion of the revised manuscript.

We also performed an additional control which is not mentioned in the manuscript. In parallel with replicating cells, we isolated the DNA from the stationary phase of growth, which primarily contains non-replicating cells. Following the three λ-exonuclease treatments, there was insufficient DNA remaining from the stationary phase cells to prepare the libraries for sequencing. This control strongly indicated that there was little to no contaminating DNA present with the SNS molecules after λ-exonuclease enrichment.

> ***Recommendations for the authors:***
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> ***Reviewer #1 (Recommendations for the authors):***
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> *Four broad issues need to be addressed.*
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> *(1) The authors have attempted to test the overlap between ORC1/CDC6 (an ORC subunit) binding in the genome and SNS-seq. If there were an overlap, this would provide evidence that the SNS-seq signals represent origins. However, the analysis provided is inadequate: merely a statement that "we obtained an overlap of 4.2% between origins and ORC1/CDC6 binding sites within a window of {plus minus}2 kb and 6.2% in the window of {plus minus}3 kb". Nowhere are these data shown or properly discussed:*
>
> *a) The authors need to provide a diagram showing where in the genome the very small amount of overlapping SNS-seq and ORC1/CDC6 binding occurs, and to clearly show and state how many of the intergenic SNS-seq peaks are sites of ORC1/CDC6 binding. In the absence of such analysis, a key question is unanswered: is there any evidence of ORC1/CDC6 (or ORC more broadly) binding at the SNS-seq signals within the polycistronic transcription units?*

In the original version of the manuscript, these data were already presented as percentages in the text and as a metaplot (Supplementary Fig. 8C).

We based our analysis on the set of 350 TbORC1/CDC6 binding sites available on TriTrypDB at the time of analysis. This dataset was a filtered subset of the originally reported TbORC1/CDC6 ChIP-on-chip peaks (personal communication, TriTrypDB). Since then, the unfiltered dataset has been made available. We therefore re-analyzed the overlap using this dataset, to which we applied a filtering that yielded 990 binding sites closely matching the 953 sites reported by the McCulloch group. We need to stress here that the original 953 sites reported by the

McCulloch group (Tiengwe et al., 2012 PMID: 22840408), is not available anymore and that the authors:

- do not provide genomic coordinates for the 953 binding sites and

- do not release any scripts or methodology that would allow independent reproduction of the 953 sites.

A similar remark also applies to the MFA-seq data (see below).

To address the reviewer's request, we have now:

(1) Recalculated the overlap using the updated TbORC1/CDC6 dataset (990 binding sites) from TriTrypDB.

(2) Added the absolute number of overlapping SNS-seq origins and TbORC1/CDC6 binding sites in the Results section for clarity.

(3) Included the TbORC1/CDC6 binding sites in the chromosomal overview (newly added to Supplementary Fig. 8A), so that their genomic localization relative to SNS-seq peaks is visually accessible.

(4) Revised the metaplots of TbORC1/CDC6 distribution around SNS-seq origins using the updated dataset (Supplementary Fig. 8C).

With these improvements, we now find that:

- Within ±2 kb, 12.9% (253) of SNS-seq origins overlap with 25.6% of TbORC1/CDC6 binding sites.

- Within ±3 kb, 18.8% (370) of SNS-seq origins overlap with 37.4% of TbORC1/CDC6 binding sites.

The updated metaplot shows a clear depletion of TbORC1/CDC6 signal at the origin center, with modest enrichment ~5 kb upstream and downstream. The underlying reason for this pattern remains unknown, and we agree that additional studies will be needed to understand it.

> *b) Equally, the authors need to explain what they conclude from this analysis. They make a comparison with T. cruzi ORC1/CDC6 and SNS-seq overlap, which does not illuminate what the data tell us. For instance, if there is no or minimal overlap between ORC1/CDC6 binding and SNS-seq peaks within the polycistronic transcription units, do they conclude that the major SNS-seq signal they detail is evidence for ORC-independent DNA replication? If there is no overlap, what further evidence can they provide that these signals truly are origins?*

First, we would like to clarify that, to date, there is no evidence supporting ORC-independent DNA replication in *T. brucei*, and—importantly—no published data demonstrating that TbORC1/CDC6 is universally required for DNA replication initiation. Because of this, we consider that it would be inappropriate to conclude that regions lacking detectable TbORC1/CDC6 signal undergo ORC-independent initiation. We would prefer not to speculate in the absence of supporting evidence and would gratefully consider any reference the reviewer wishes to provide on this subject.

Second, the low overlap between TbORC1/CDC6 binding sites and SNS-seq origins does not, in our view, invalidate our mapping of replication initiation sites. Multiple factors contribute to this:

(1) Low overlap between ORC1/CDC6 and origin-mapping techniques has been repeatedly reported across kinetoplastids. For instance, in *T. cruzi*, 88.2% of origins detected by DNAscent nanopore sequencing showed no overlap with TcORC1/CDC6–Ty1 ChIP signal within ±3 kb, and only 11.7% co-localized. This is strikingly similar to our observations in *T. brucei.* Thus, our data are consistent with the broader pattern in trypanosomatids rather than an exception.

(2) The origin topology detected by stranded SNS-seq is supported by several genomic characteristic found frequently in other eukaryotes, including:

- A highly specific and polarized poly(dA)/poly(dT) sequence environment.

- Strand-specific G4 structures positioned around origin centers.

- A conserved nucleosome-depleted region flanked by well-positioned nucleosomes.

These features are absent from shuffled controls, appear at high significance, and recapitulate hallmark signatures of replication origins in other eukaryotes.

Together, these findings give us confidence that the SNS-seq peaks represent genuine origins - despite the incomplete overlap with TbORC1/CDC6 binding.

Third, we fully agree with the reviewer that a definitive conclusion would require an additional, independent validation method.

Given the lack of complete ORC subunit datasets and the unusual biology of trypanosomatid replication complexes, we believe that the cautious interpretation above is the most appropriate.

> *c) The authors state (Discussion): "Validation of origins is generally a difficult task, particularly in trypanosomatids, where proteins involved in the initiation of DNA replication are difficult to determine. Few proteins have been described as potential ORC subunits (reviewed in 61), and none of them have been shown to be a specific marker that indicates the origins." There are two problems with the statement. First, most of the subunits of ORC have now been described in T. brucei; the authors should make this clear. Second, mapping of ORC1/CDC6 localisation, contrary to what the authors state here, shows precise correlation with the peaks of every MFA-seq signal described (see Tiengwe et al, Cell Reports, 2012); thus, ORC1/CDC6 binding provides evidence that MFA-seq is detecting origins, something that cannot be said for SNS-seq. The authors need to correct this misleading paragraph.*

As suggested, we have removed the paragraph from the Discussion to avoid confusion. However, we disagree with the reviewer's assessment and clarify below our position regarding the issues raised.

First, we agree that five candidate ORC subunits have now been identified in *T. brucei*. Our intention was not to suggest the contrary, but rather to emphasize that, although candidate ORC components have been described, direct functional evidence for their roles in replication initiation is still limited. For this reason, we were cautious in referring to any ORC component as a definitive marker of replication origins.

Second, regarding the reviewer's statement that TbORC1/CDC6 binding "shows precise correlation with the peaks of every MFA-seq signal", we respectfully disagree based on several observations:

(1) MFA-seq does not identify individual origin centers, but rather broad replicated regions that often span hundreds of kilobases. By design, this method cannot define the number or

position of discrete origins within each peak. For that reason, MFA-seq regions do not have the resolution required to validate TbORC1/CDC6 binding sites as individual origins.

(2) In the published datasets (Tiengwe et al., Devlin et al.), no metaplots or locus-wide quantification of the overlap between MFA-seq peaks and TbORC1/CDC6 binding were provided. The coordinates or the approach used to define the discrete regions that they define as the originsin the MFA-seq broad peaks have never been described or made available, making it difficult to evaluate the claimed correspondence.

(3) Notably, McCulloch's group later reported that only 4.4% of the 953 TbORC1/CDC6 sites overlapped with their 42 MFA-seq "origins", underscoring that the degree of correspondence is in fact limited (PMID: 29491738).

(4) Finally, as noted in our response to point (1b), low overlap between ORC1/CDC6 binding sites and origin-mapping techniques is a consistent observation across kinetoplastids, including *T. cruzi*, where DNAscent-mapped origins show only ~12% overlap with TcORC1/CDC6 ChIP signals. This suggests that the limited overlap we observe is not unique to our dataset.

For these reasons, we are not convinced that the TbORC1/CDC6 binding sites have been shown to align precisely with MFA seq peaks, nor that these datasets definitively validate origin mapping in *T. brucei*. Nevertheless, to avoid over-interpretation and potential confusion, we have removed the paragraph from the Discussion as requested. We hope this clarifies our position and improves the accuracy and neutrality of the manuscript.

> *(2) Like for ORC1/CDC6 localisation, the authors' evaluation of the relationship between MFA-seq and SNS-seq mapping is inadequate, and the depth of the analysis and discussion needs to be improved:*
>
> *a) The authors state: "We found 28-42% stranded SNS-seq origins overlapped with early and 43-55% overlapped with late S-phase MFA-seq replicated regions (Supplementary Figure 8B)." This seems important and provides (limited) validation of both datasets, but cannot be discerned from the supplied figure. Please provide a metaplot of the two datasets centred on the MFA-seq loci, including the SNS-seq peak amplitude.*

We would like to emphasize that MFA-seq is not a method designed to map individual origins, and this fundamentally limits the interpretability of metaplots centered on MFA-seq regions. MFA-seq identifies broad replication-enriched domains, typically spanning 100–500 kb, within which multiple origins may fire asynchronously across the cell population.

This concern is reinforced by the original MFA-seq publications (Tiengwe et al., 2012; Devlin et al., 2016), which:

- do not provide positional data for the 42-47 MFA-inferred origins,

- do not describe the computational method used to derive individual origin coordinates from the broad peaks, and

- do not release any scripts or methodology that would allow independent reproduction of the claimed origin positions.

Because of this, it is not possible to reconstruct or validate how the 42 MFA-seq "origin" sites were defined, nor to use those coordinates as anchors for metaplot analyses.

Most importantly, we disagree with the underlying assumption that each MFA-seq peak corresponds to exactly one origin. This assumption runs counter to the principle of the technique, which identifies regions of higher DNA content in replicating cells than in non-replicating cells; it is also contradicted by our stranded SNS-seq data and by DNA combing measurements:

- SNS-seq detects multiple discrete origins within the same genomic regions that produce a single broad MFA-seq peak.

- DNA combing reveals inter-origin distances of ~36–422 kb (median ~150 kb) (PMID: 26976742), which is far shorter than the ~400–600 kb replication domains identified by MFA-seq.

- Furthermore, with only 42 origins detected by MFA-seq, it is not possible to achieve complete genome replication in *T. brucei* during S-phase. DNA combing has found that the average speed of replication forks in the procyclic forms is 1.9 Kb/min. (PMID: 26976742). Dividing the size of the *Trypanosoma brucei brucei* TREU927 genome (26.1 Mb) by 42 origins (PMID: 22840408) shows that 621 Kb must be replicated during the S phase. Using the calculated average replication speed of 1.9 Kb/min, we can estimate that the replication of 621 Kb would take 327 min (5.45 hours) (621 Kb/1.9 Kb/min = 327 min). However, this exceeds the estimated length of the S-phase in these parasites, which is 2.31 hours (138.6 minutes) (PMID: 32397111, 31811174, 28258618) or less, 1.36 hours (PMID: 2190996, 10574712) in *Trypanosoma brucei* procyclic forms. Therefore, more than 42 origins are necessary to complete replication during the short S phase.

This makes it unlikely that MFA-seq regions represent single functional origins. For these reasons, a metaplot centered on MFA-seq "loci" may lead to misinterpretations and would not provide biologically meaningful information.

We hope that the expanded explanation clarifies our interpretation of the relationship between these two complementary, but fundamentally different, methods.

> *b) The authors state that "Our results showed that the origins are predominantly located in the intergenic regions within the PTUs (Figure 2C)'. This finding cannot be discerned from this figure, which does not show 'strand switch regions' (SSRs; transcription start/stop sites), where MFA-seq predicts all origins to localise. The authors need to acknowledge this difference and must show a comparison of SNS-seq data, including peak amplitude, around all SSRs (whether predicted by MFA-seq to act as origins or not, since all appear to bind ORC1/CDC6).*

We have now provided the metaplots showing the overlap between stranded SNS-seq origins and SSRs (see Supplementary Figure 8D). This difference has been acknowledged and discussed in the revised manuscript.

> *c) Finally, the authors' interpretation that around 30-55% of SNS-seq peaks overlap with MFA-seq 'origins' is highly questionable. MFA-seq peaks are regions of increased DNA content in replicating cells relative to non-replicating cells, and so the entire region under the MFA-seq peak is not necessarily an origin, but is likely to be a more discrete locus (eg, the SSR, where ORC1/CDC6 mainly localises). They should correct the wording and discuss what significance they see in this overlap; for instance, do they think SNS-seq 'clusters' are more pronounced within the MFA-seq peaks and, if so, what might this mean, and why does it not correlate with ORC1/CDC6 localisation?*

As the reviewer notes, 'MFA-seq peaks are regions of increased DNA content, and so the entire region under the MFA-seq peak is not necessarily an origin but is likely to be a more discrete locus'. This is exactly why MFA-seq is inappropriate for identifying discrete/individual origins: within these replicated domains, multiple origins can fire, as revealed both by stranded SNS-seq mapping.

Regarding the overlap between SNS-seq origins and MFA-seq peaks, we agree with the reviewer that this overlap should not be interpreted as validating MFA-seq "origin positions." Instead, we now describe it more accurately as the proportion of discrete SNS-seq origins that fall within broader MFA-seq replication domains. This is expected, because SNS-seq identifies individual initiation events, whereas MFA-seq identifies S-phase replication domains averaged across a population. Our stranded SNS-seq data do not show enhanced origin accumulation within MFA-seq regions, and we find no correlation with TbORC1/CDC6 positions. This is now discussed.

Regarding SSRs, we do not share the view that they should be considered privileged initiation sites. After remapping the TbORC1/CDC6 ChIP-on-chip dataset (see above) to the *T. brucei* Lister 427–2018 genome (Supplementary Fig. 8A), we observed that TbORC1/CDC6 binding is distributed throughout the chromosomes, not restricted to SSRs. To quantify this, we analyzed the overlap between TbORC1/CDC6 sites and all annotated SSR classes (dSSRs, cSSRs, and head-to-tail regions, as defined in Kim et al. 2009). The results show that:

Only 10% of TbORC1/CDC6 binding sites fall within 40% of all SSRs.

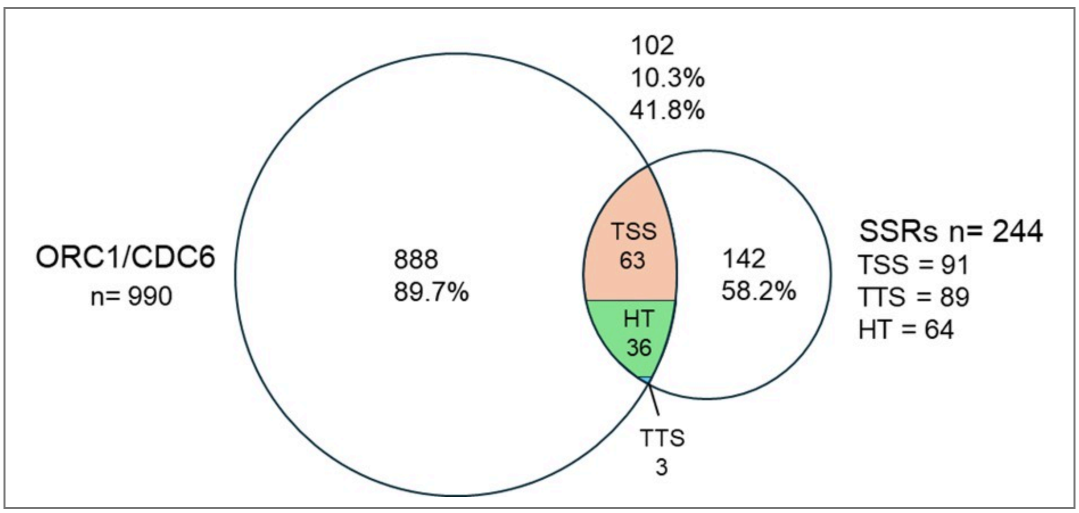
At the level of individual SSR types:

- TTS: 3.3% of TTS overlap with 0.3% of TbORC1/CDC6 sites.

- TSS: 67% of TSS overlap with 6.1% of TbORC1/CDC6 sites.

- Head-to-tail regions: 54.2% overlap with 3.6% of TbORC1/CDC6 sites.

These analyses demonstrate that most TbORC1/CDC6 sites are *not* located at SSRs, contradicting the idea that SSRs represent primary or exclusive origin sites.



Author response image 1. Overlap between TbORC1/CDC6-12Myc binding sites (Tiengwe 2012, Cell Reports) and strand-switch regions (SSRs). Venn diagram showing the overlap of 990TbORC1/CDC6-12Mycbinding sites (Retrieved from TritrypDB filtered at score 22 to achieve a number of binding sites similar to the one (953 binding sites) published in Tiengwe 2012, Cell Reports) and SSR sites in the genome (Kim 2018, NAR). The intersection shows that 10.3% of Orc1/CDC6 binding sites overlap with 41.8% SSRs. The intersection is subdivided into TSS (orange), TTS in (blue) and HT in (green).

> *(3) A key objection to the data presentation is the decision to limit SNS-seq mapping to the intergenic regions. In addition to overlooking the SSRs (see above, 2), so-called subtelomeres, which account for nearly 50% of the T. brucei genome and are largely*

> *untranscribed, are not shown or discussed at all. Providing this data will improve clarity and also provide a key test of one of the predictions that the authors make: "most origins are localized in actively transcribed regions, which could lead to collisions between DNA replication and the transcription machinery. This spatial coincidence implies that transcription and replication must occur in a highly ordered and cooperative manner in T. brucei."*

We do not understand why this reviewer concluded that we took 'the decision to limit the mapping of SNS-seq to intergenic regions'. This is a factual error.

To be clearer,

(2) We now explicitly present the distribution of SNS-seq origins across core and subtelomeric regions in the revised Figure 2D, making clear that origin mapping was performed genome-wide.

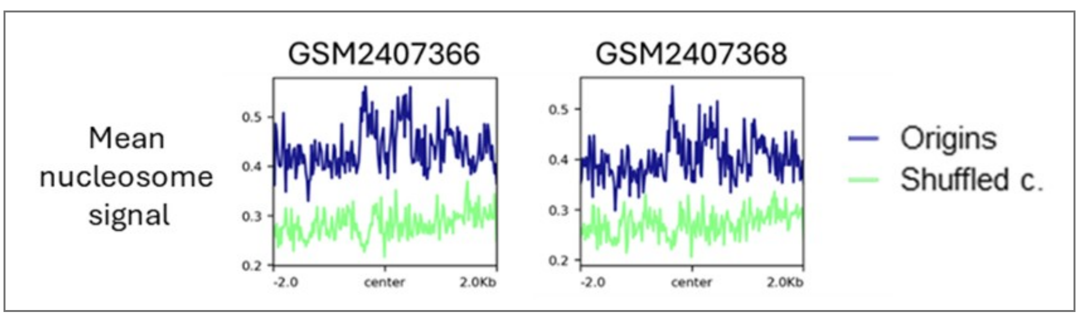
(2) And that SNS-seq origins are also present in subtelomeric regions. We have revised the manuscript to avoid any implication that origin firing is restricted only to actively transcribed regions. Our data show that most SNS-seq origins lie within intergenic regions of PTUs, but a minority are found outside these regions—including subtelomeres and SSRs. The revised text reflects this nuance and highlights that the spatial relationship between transcription and replication is strong but not exclusive.

These additions undoubtedly ensure that the genomic-wide nature of SNS-seq analysis is transparent to the reader and should therefore remove this reviewer's “key objection”.

> *a) The authors must show SNS-seq mapping to the subtelomeres (in addition to around the SSRs; see comment (2). If no SNS-seq peaks are detected in the subtelomeres, what do the authors conclude about how the genome is duplicated? If SNS-seq peaks are detected in the subtelomeres, do they correspond with the ordered nucleosomes in this part of the genome described by Maree et al (PMID: 28344657); if so, might SNS-seq signal localisation not be directed by transcription but chromatin?*

We have now presented the proportion of origins in subtelomeric regions (see Figure 2B).

As illustrated in the metaplots in Author response image 2, the distribution of nucleosomes around the subtelomeric origins is similar to the distribution shown for all origins in the manuscript. We do not see the pattern of nucleosomes as described by Maree et al (PMID: 28344657) over ORC1/CDC6 binding sites in this part of the genome.



Author response image 2. Metaplots showing the mean nuclesome signal over centred SNS-seq origins in subtelomeric regions. Two replicates from Maree et al 2019 (PMID: 28344657).

We never claimed that transcription directs the localisation of the SNS-seq signal. We did not conduct experiments to address this issue. In contrast, we consider that the organisation of chromatin exerts a significant influence on the selection of active origins.

> *(4) The major conclusion of the manuscript is that the SNS-seq signal corresponds very precisely to the locations of RNA-DNA hybrids (R-loops). Given all the limitations discussed above, can the authors rule out the possibility that SNS-seq is merely mapping DNA-DNA hybrids and is not, in fact, detecting origins?*
>
> *a) It is legitimate to speculate about the possibility that the very extensive overlap between SNS-seq and DRIP-seq signals within polycistronic transcription units (between ORFs) might suggest that DRIP-seq data detects nascent strands at replication origins, rather than R-loops at sites of pre-mRNA processing, as previously suggested by Briggs et al (PMID: 30304482). (eg, 'we disclosed for the first time a strong link between R-loop formation and DNA replication initiation'; 'The RNA:DNA hybrids are formed at initiation sites by RNA priming of SNS and Okazaki fragments'). However, the authors should acknowledge that alternative explanations for the localisation and potential functions of inter-CDS R-loops have been suggested,*

We do not find extensive overlap between stranded SNS-seq and DRIP-seq signal. We have observed only a minor proportion (1.7%) of the previously reported DRIP-seq signal to overlap with the origins detected by stranded SNS-seq. The RNA-primed SNS must form RNA:DNA hybrids during the initiation of DNA replication, and that an enrichment of these hybrids around the origins is expected. Therefore, we legitimately speculated that this minor proportion of RNA:DNA hybrids enriched around origin centres could be due to the origin activation.

We agree that some of the DRIP-seq signals detected around the origins may be sites of pre-mRNA processing, as previously suggested by Briggs et al. (PMID: 30304482). Since there is no data proving implication of pre-mRNA processing into DNA replication initiation we prefer not to speculate about it.

> *b) More importantly, the authors should provide experimental evidence that tests such a mechanistic prediction of R-loops and origins: for instance, have they attempted to remove R-loops, eg, by treatment with RNase H, and checked that the SNS-seq signal is unaltered? In the absence of such data, they cannot exclude the possibility that their work has revealed an overlooked problem with SNS-seq (which may not be limited to T. brucei; are matched DRIP-seq and SNS-seq datasets available to correlate these signals in a range of organisms?).*

We have not attempted RNase H treatment for a fundamental methodological reason: it seems highly improbable that RNA:DNA hybrids would persist through the multiple denaturation steps inherent to the SNS-seq enrichment protocol. Published biophysical measurements show that RNA:DNA hybrids melt at ~95 °C (Roberts & Crothers, *Science*, 1992; PMID: 1279808), which is the temperature repeatedly applied during SNS isolation. Under these conditions, persistent RNA:DNA hybrids cannot remain intact and therefore cannot be responsible for the SNS-seq peaks detected.

We do not interpret our findings as revealing an “overlooked problem with SNS-seq.” Instead, we consider that the enrichment of RNA:DNA hybrids around origins observed in DRIP-seq is biologically meaningful and expected, given that replication initiation involves RNA-primed nascent strands and that DRIP-seq detects such structures.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *I have some minor concerns that do not affect the main conclusions of the manuscript:*
>
> *(1) Figure 2B: The regions shown in the heatmap have different sizes, and I presume that the regions are ordered by size on the y-axis? If so, does the cone-shaped pattern, which is origin-less for genic regions and origin-enriched for intergenic regions, arise from the*

> *size of the regions? (I.e., for each genic region, the region itself is origin-less and the flanking intergenic regions contain origins.) If this is the case, then the peaks/valleys, centered exactly on the center of the regions on the mean frequency plots, arise from the different sizes of the analyzed regions, not from the fact that origins are mostly found at the center of intergenic regions.*

That is correct. The regions displayed in the heatmaps are genic and intergenic region sorted by size. We did not want to convey with this metaplot that the origins are accumulating at the centres of the intergenic region but mainly that genic regions are mostly devoid of origins and the intergenic regions enriched in origins.

> *(2) Line 123, "and the average length of origins was found to be approximately 150 bp.": To determine origins, the authors filter away overlapping peaks and peaks that are too far from each other. Both restrict the minimal and maximal length of origins that can be observed, and this, in turn, affects the average length.*

This observation is correct. By applying filtering and setting the maximum distance between the positive and negative peaks, we are most likely affecting the average length by excluding origins that are potentially wider. Nevertheless, the violin plot shows that the majority of origins are shorter than 500 nt. In the end, the size of regions detected as the origin is not important. What gives the resolution of stranded-SNS-seq is the ability to identify the centre of the origin between the minus and plus peaks.

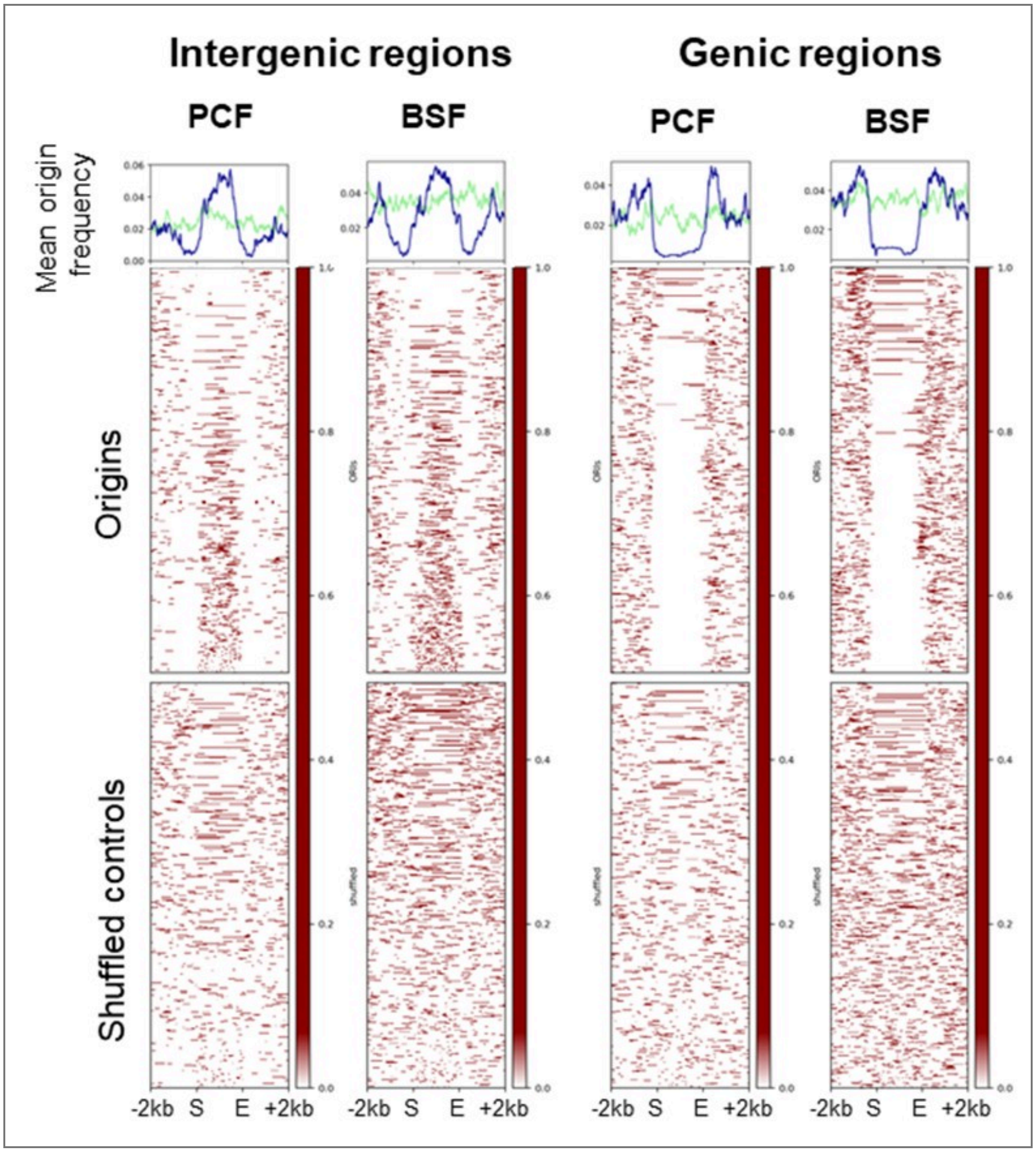
> *(3) Data in the manuscript were sometimes not presented in an easy-to-read manner. In some cases, this was due to benign things, such as missing labels for the mean frequency plots (e.g., Figure 2B, blue and green) or very small fonts for axes (Figure 2B). Sometimes, due to the plot types that were chosen, such as pie-charts (Figure 2C, see https://medium.com/analytics-vidhya/dont-use-pie-charts-in-data-analysis-6c005723e657), stacked bar plots (Figure 6B), or showing cumulative distributions (Figure 5C, and Figure 2D) it makes it difficult to judge the actual distribution.*

Wherever possible, the size of the small fonts was increased to the maximum. Missing labels were added to the mean frequency plots. We increased the font size for the axes in the frequency plots.

However, we found cumulative distributions useful. If you have a more specific proposal for replacing cumulative distributions, we would be very grateful to hear it. We also hope that magnifying the figures in TIFF format with a higher resolution will improve visibility.

> *(4) Figure 2B: This data would be better presented with all regions stretched to the same size (the reason is explained in the public review).*

We performed the scaled plots for the stranded SNS-seq origins over the genic and intergenic regions as the reviewer suggested (see Author response image 3), but we prefer to keep the unscaled versions in the manuscript.



Author response image 3. Distribution of mapped origins in scaled genic and intergenic regions. Scaled heatmaps present the distribution of the mapped origins and shuffled controls within scaled genic and intergenic regions (± 2 kb).

> *(5) Line 149: "The number of origins in both cells was 148 compared using normalised mapped reads": Supplementary Figure 2D mentions that conditions were subsampled to the same amount. I would mention that explicitly in the main text ("compared using normalized, subsampled mapped reads"), as 'normalizing' would not include 'subsampling' for me. Also, I could not find the methods section that the authors refer to here.*

Thanks for the suggestion. We changed the text to make this point clearer. In the methods section, the subsampling process was referred to as 'PCF down-sampling', but we changed now the name to 'Read sub-sampling' to be more consistent in the edited version of the manuscript.

> *(6) Figure 2C: I struggled to understand what gDNA stands for. Maybe it could be replaced with something like distribution in genome?*

Thanks for this suggestion. It is changed to ‘distribution in genomic sequence’.

> *(7) Figure 5C: I cannot see how a G4 30 kb from an origin could be relevant. This also does not fit the scale of the author's own model at all (Figure 8).*

The main goal of Figure 5C was to demonstrate the differences between origins and the nearest G4s compared to the shuffled controls. The graph shows that 50% of the origins have a G4 within 2010 bp, whereas the median for the shuffled control is 4154 bp in the case of non-stabilised G4s. Our model is based on Figure 5D, which illustrates the enrichment of G4s and poly(dA) around the centre of origins.

> *(8) Figure 6B: could be made supplementary in my opinion. All relevant data is repeated in panel D.*

It is true that Figures 6B and 6C contain some repetition. However, we would prefer to keep Figure 6B because it provides a quantification of the six indicated categories, along with the statistical tests. Figure 6B only presents the three categories that changed significantly. Figure 6D shows distribution but does not contain quantified data.

> *(9) Figure 6D: This plot is repeating a lot, within single figures (Figure 6A, top) but also between figures (e.g., Figure 5D, Figure 4B). I'd prefer it if the initial plots of each figure were expanded a bit (here Figure 6A, top) to include some information from the previous figures. Then all these summary plots could be combined into a single figure at the very end (maybe still as different panels to reduce the number of lines in a single plot). Otherwise, each summary plot repeats the tracks of the previous, which becomes very repetitive.*

Our model is based on these summary plots, and we calculated the relative distances between the different elements using them. Two elements were repeated in each plot: the positions of poly(dA) and G4s. These two elements served as reference points to determine the relative positions of the other elements. Following your suggestion would result again in repetitive summary plots at the end, as one combined summary plot would be overloaded with lines and difficult to understand.

> *(10) Figure 6D & Figure 7C: Both show predicted G4s; however, on the plus strand, one prediction has a two-peaked shape, the other only a single peak. Is this a mistake?*

The graphs for the predicted G4s do not have the same shape in the two plots as they were performed in different reference genomes for *T. brucei*. Figure 6C is in the 427-reference genome as the MNase-seq data set was analysed in this reference genome and we re-did the SNS-seq analysis and the G4 prediction in this reference genome to be able to compare them directly. In Figure 7C we are comparing origins DRIP-seq and predicted G4s, in this case all datasets could be compared in the 427-2018 reference genome.

https://doi.org/10.7554/eLife.108143.2.sa0